

gravitational waves

observing black-hole

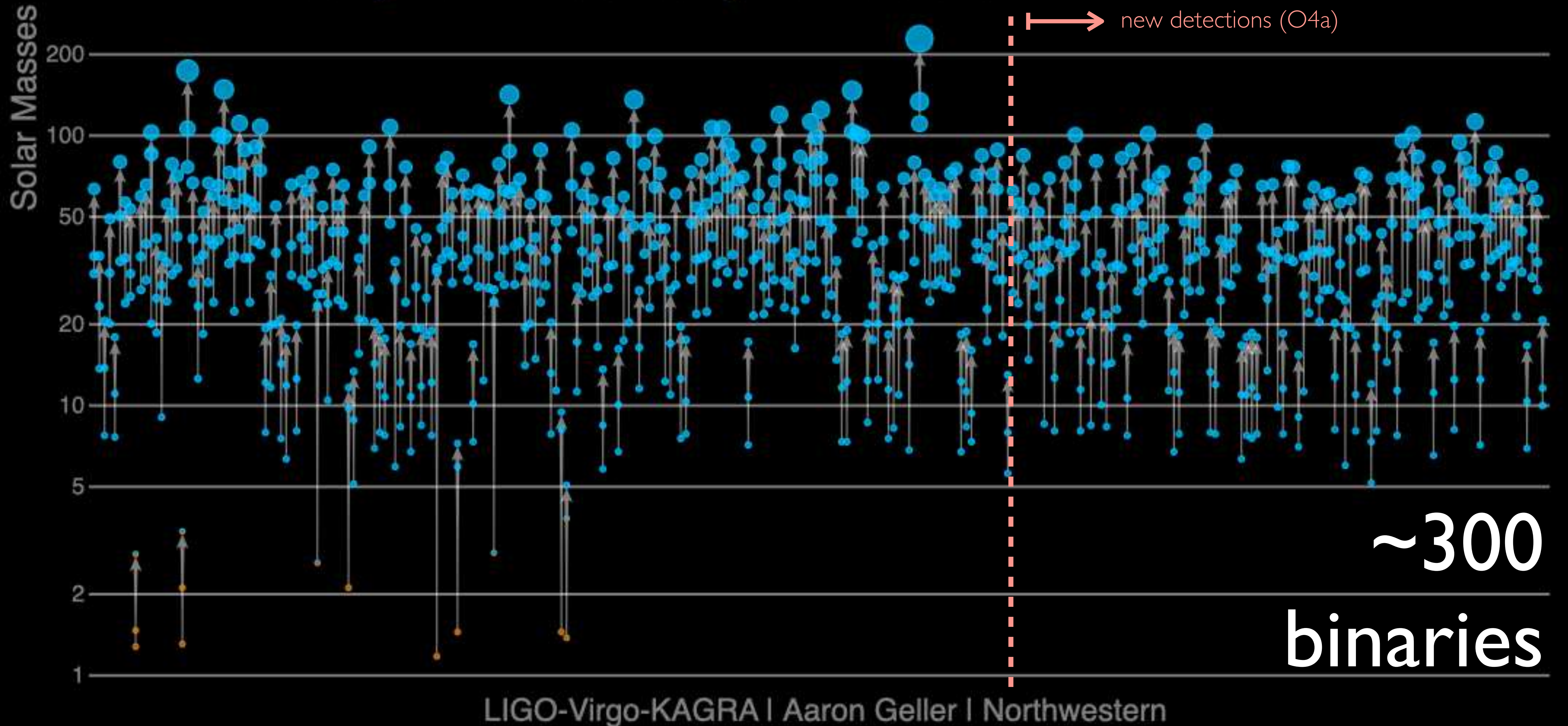
ringdowns



Maximiliano Isi | Columbia University & Flatiron Institute

Masses in the Stellar Graveyard

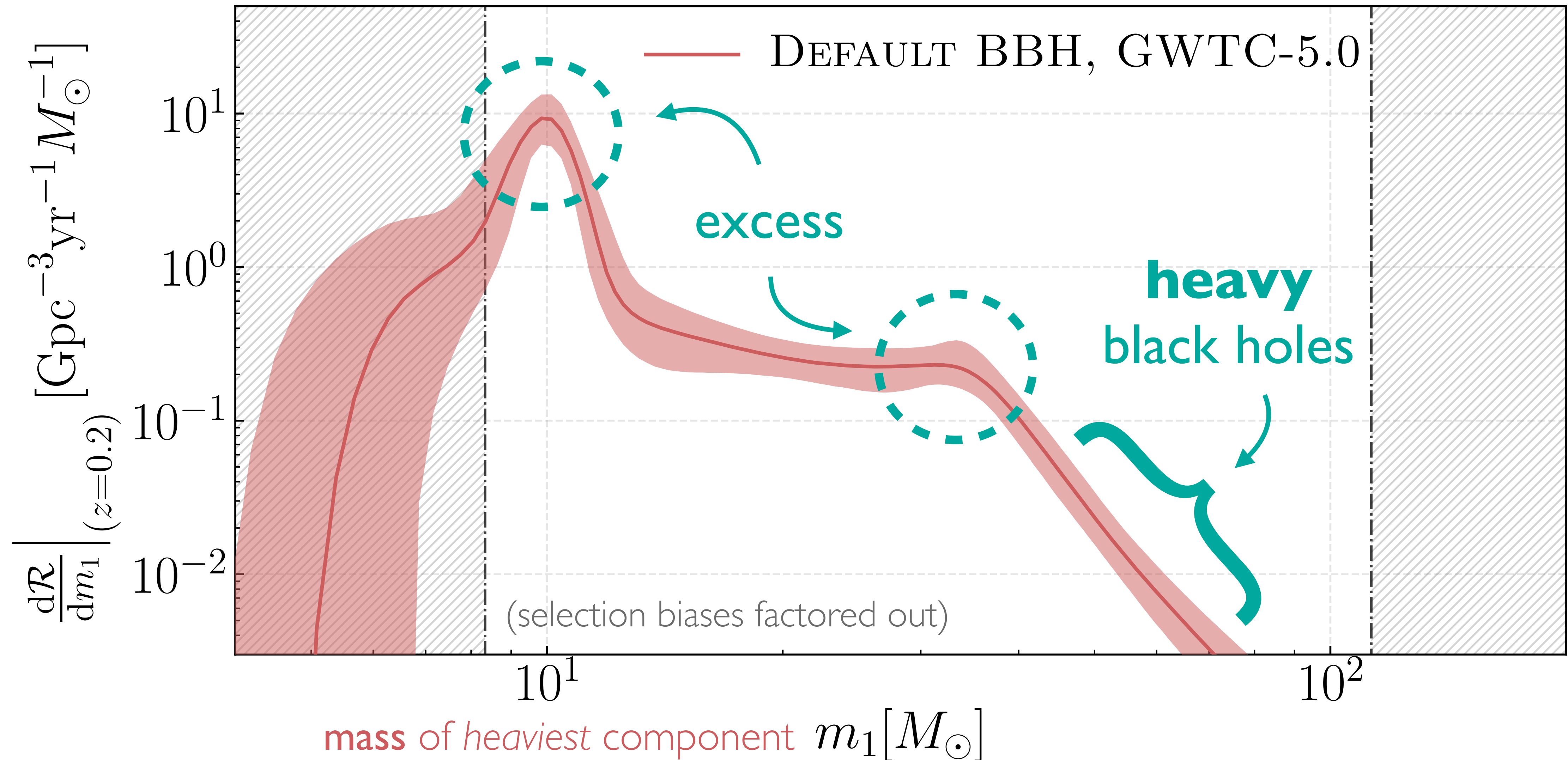
LIGO-Virgo-KAGRA Black Holes LIGO-Virgo-KAGRA Neutron Stars GWTC-5



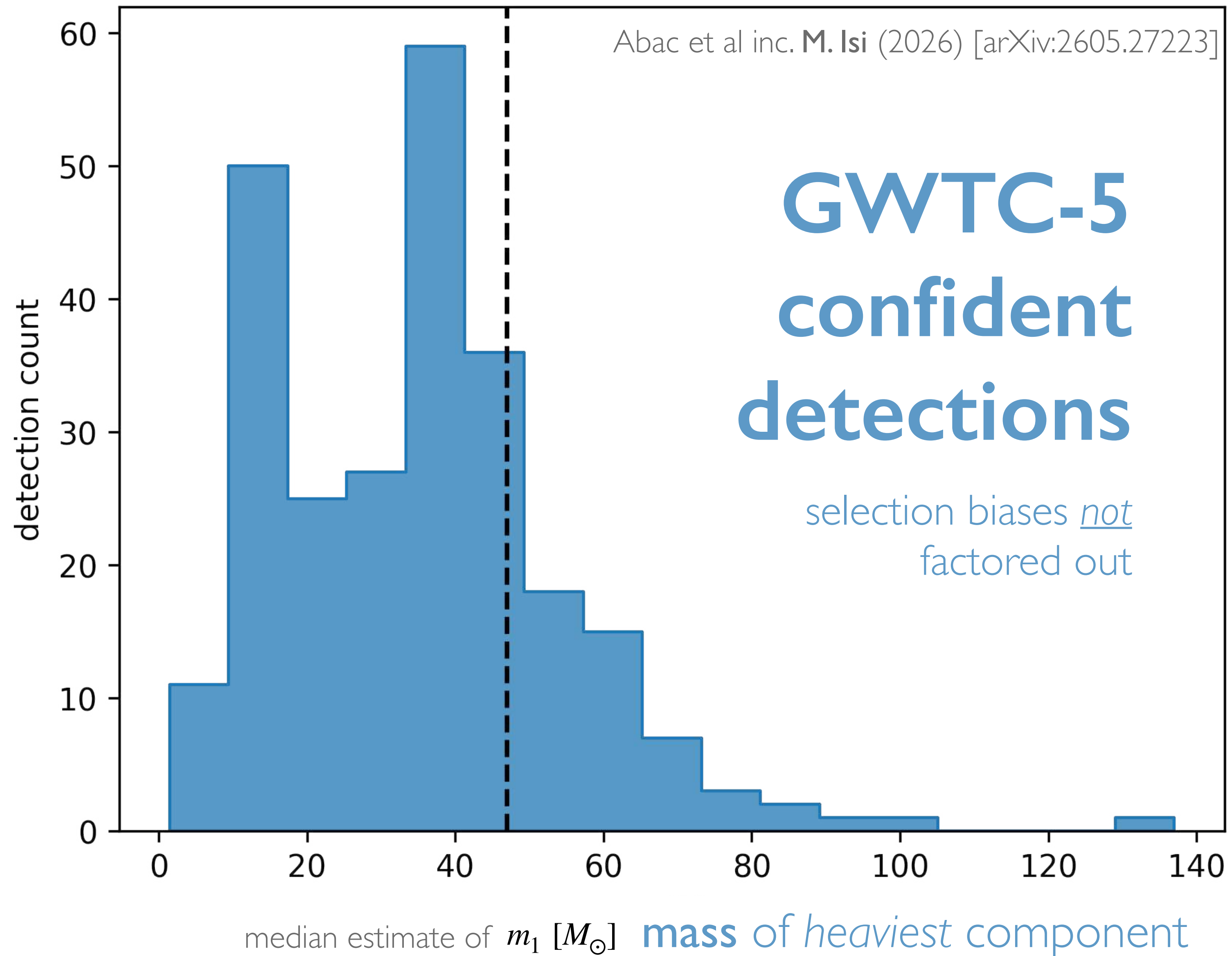
black-hole *mass* distribution

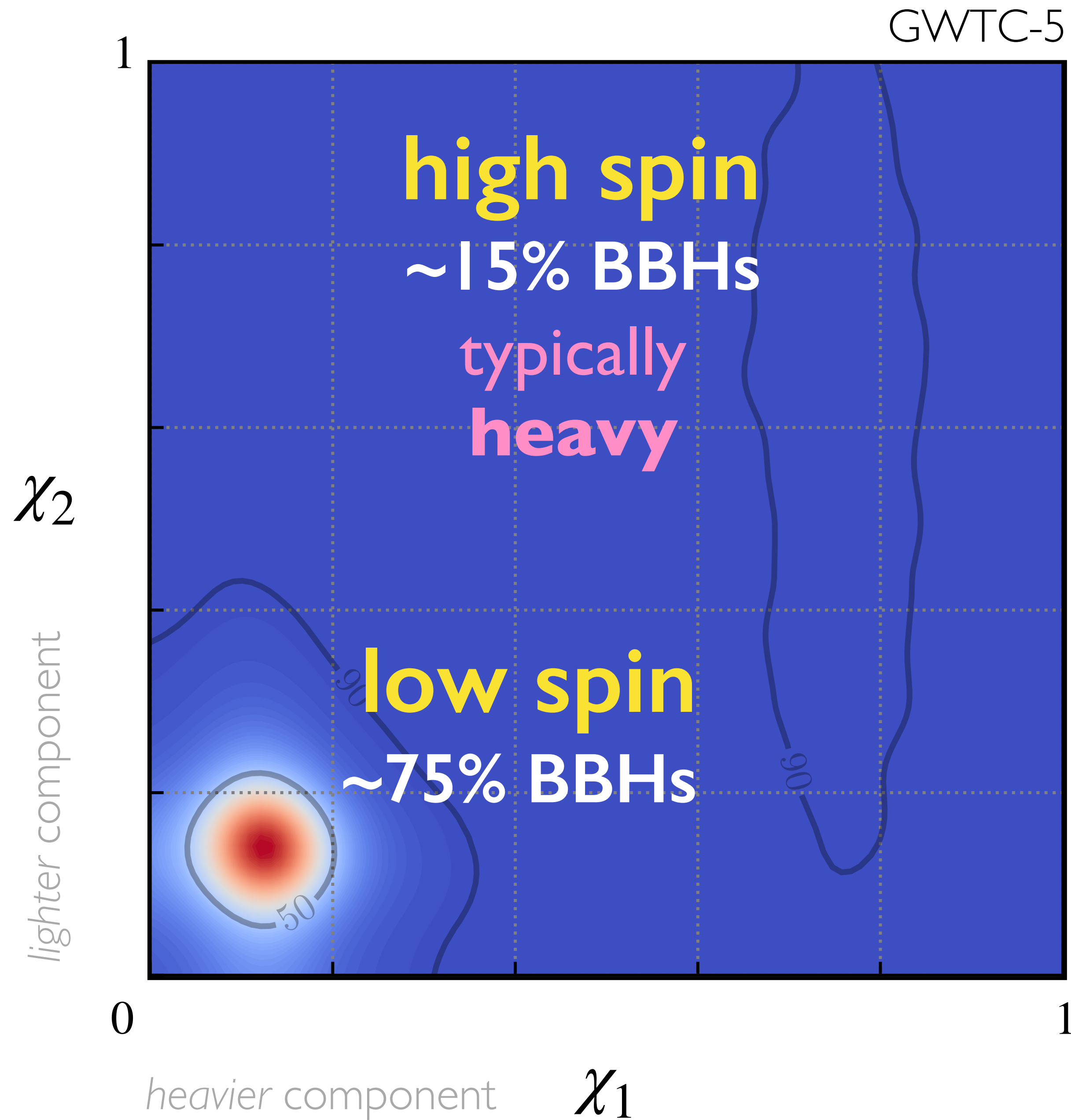
LIGO-Virgo-KAGRA (latest estimate)

Abac et al inc. **M. Isi** (2026) [arXiv:2605.27226]



detected black-hole masses





distribution of BH spin magnitudes

2 subpopulations

high & low spin

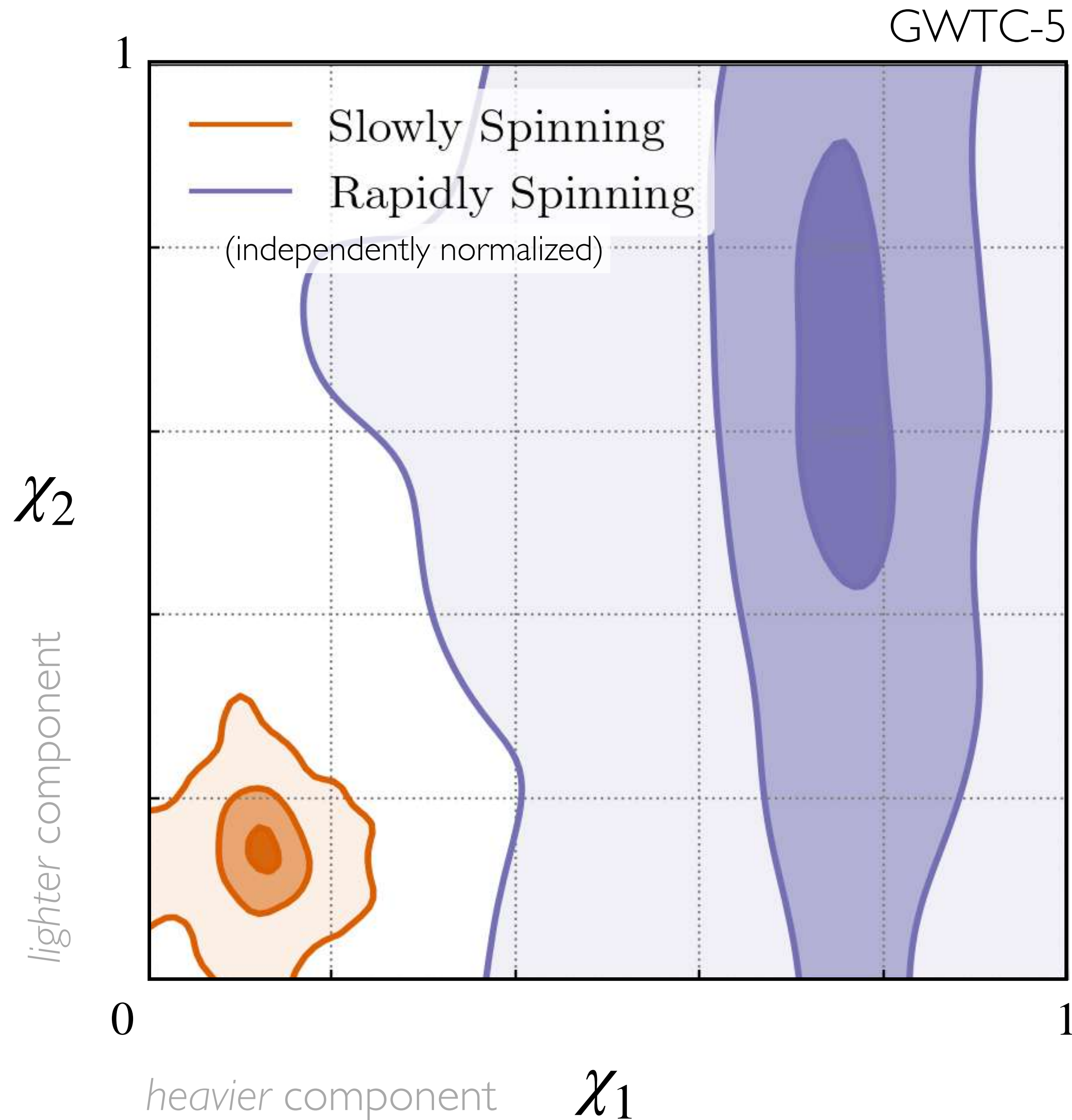
heavier $\rightarrow$ fast spin



A. Hussain*, M. Isi, A. Zimmerman
[arXiv:2411.02252, arXiv:2605.24281]

*postdoc at CCA

Abac et al inc. M. Isi (2026) [arXiv:2605.27226]



distribution of BH spin magnitudes

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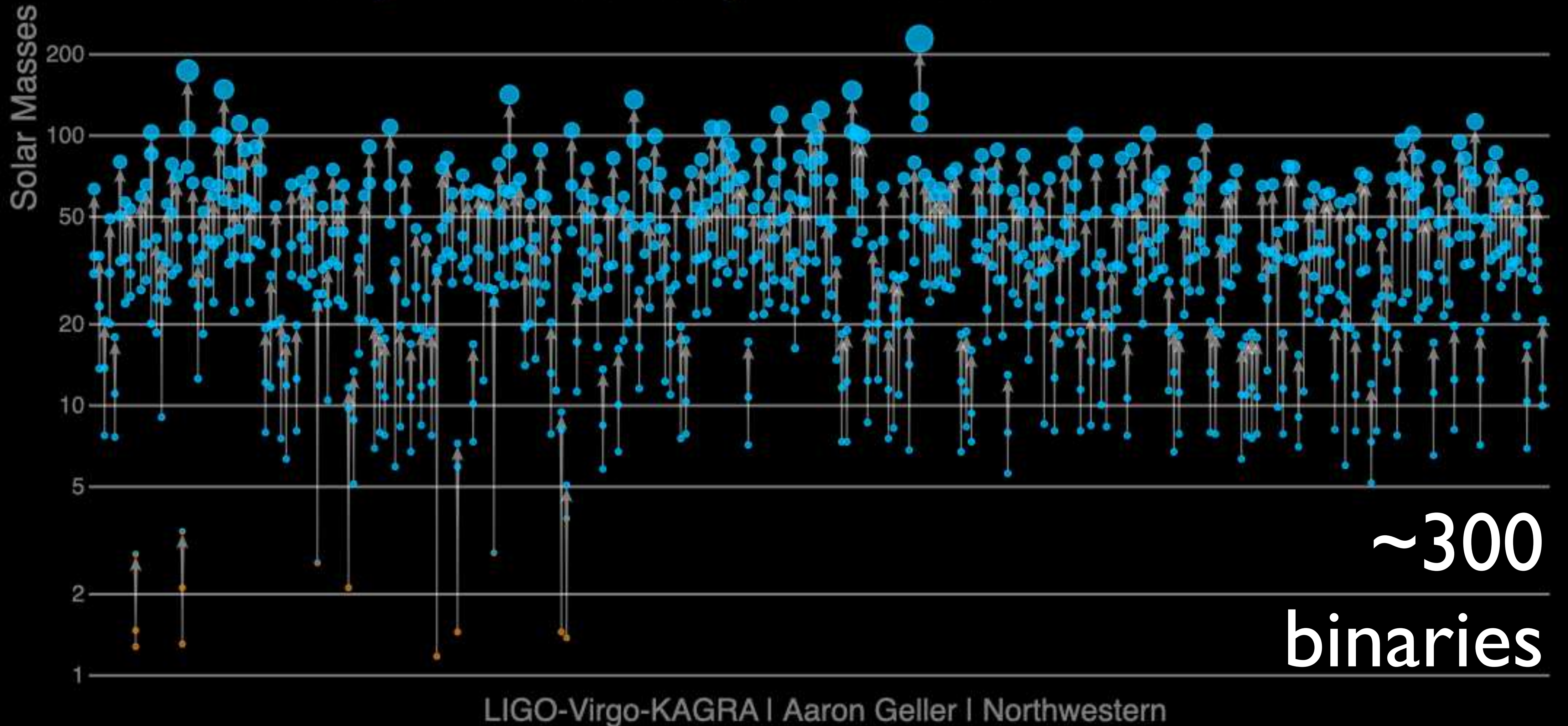
A. Hussain*, **M. Isi**, A. Zimmerman
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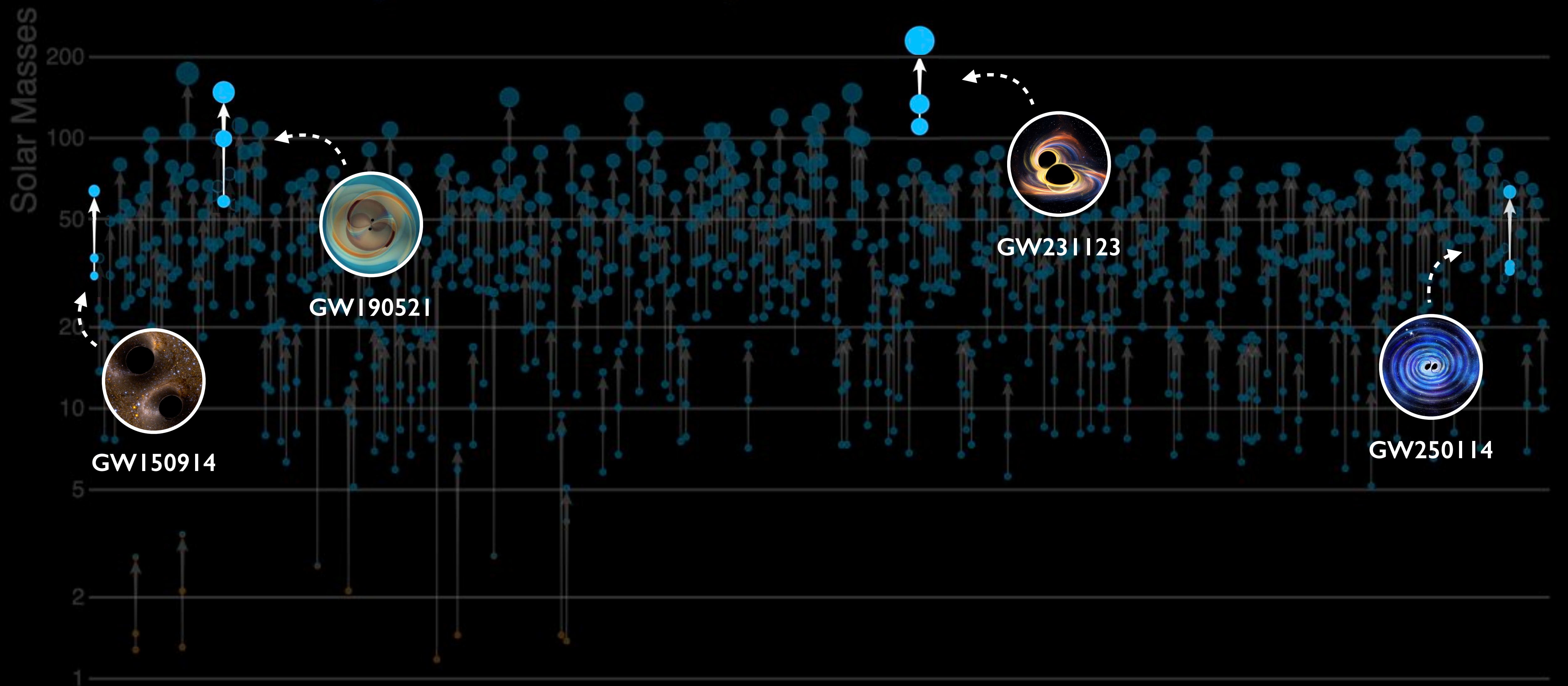
Masses in the Stellar Graveyard

LIGO-Virgo-KAGRA Black Holes *LIGO-Virgo-KAGRA Neutron Stars* **GWTC-5**



Masses in the Stellar Graveyard

LIGO-Virgo-KAGRA Black Holes LIGO-Virgo-KAGRA Neutron Stars GWTC-5

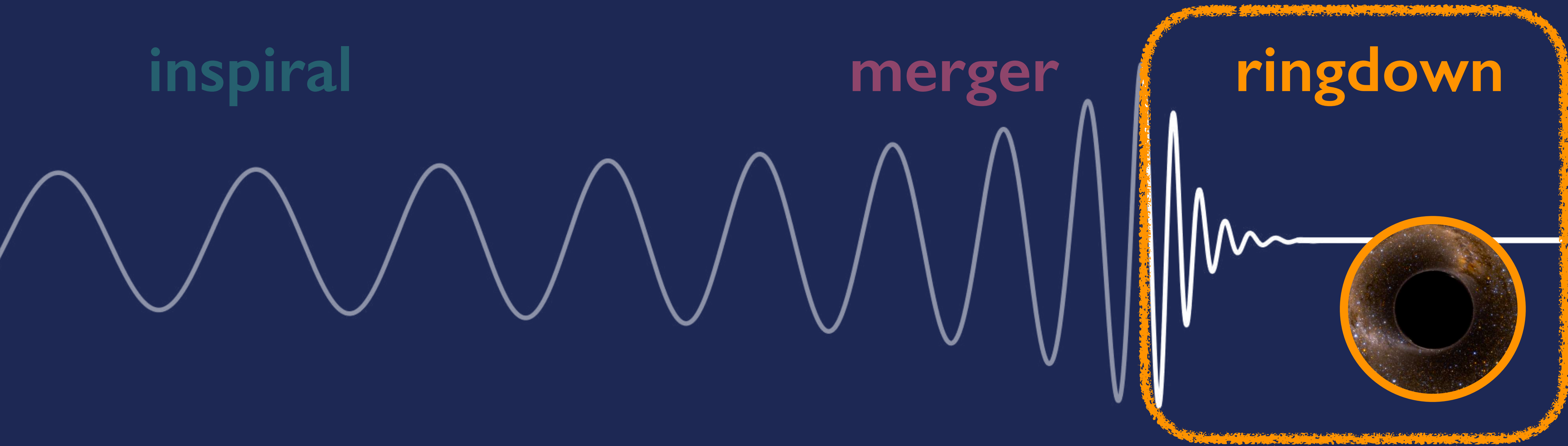


LIGO-Virgo-KAGRA | Aaron Geller | Northwestern

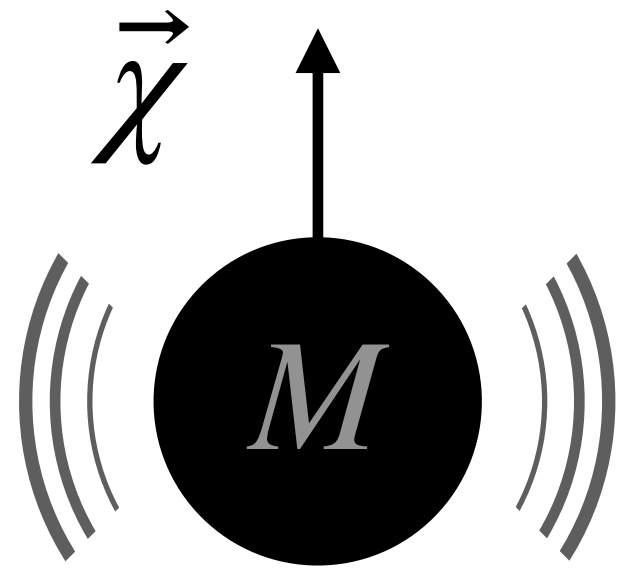
inspiral

merger

ringdown



ringdowns 101



ringdown

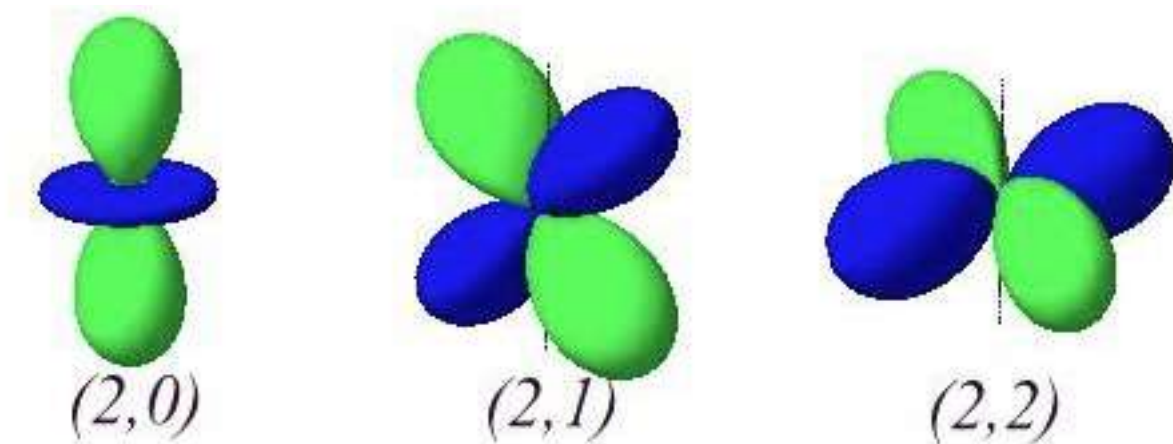
$$h \equiv h_+ - ih_\times$$

$$h = \sum_{\ell \geq 2} \sum_{|m| \leq \ell} \sum_{n \geq 0} \left[C_{\ell mn} e^{-i\tilde{\omega}_{\ell mn} t} - 2S_{\ell m}(\tilde{\omega}_{\ell mn}, \chi; \iota, \varphi) \right]$$

$\omega_{\ell mn} - i/\tau_{\ell mn}$

overtone number

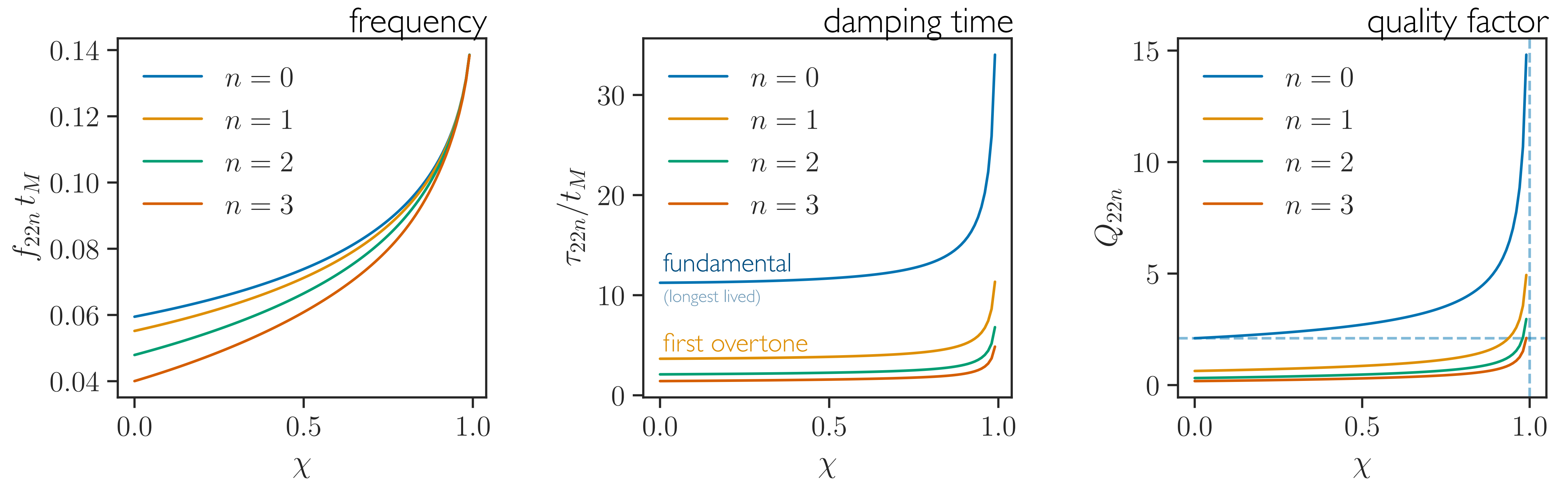
$$\tau_{\ell mn} \geq \tau_{\ell m(n+1)}$$



$$-2S_{\ell m}(\tilde{\omega}_{\ell mn}, \chi; \iota, \varphi) \approx -2Y_{\ell m}(\iota, \varphi)$$

overtones

infinite family of tones with a given ℓm

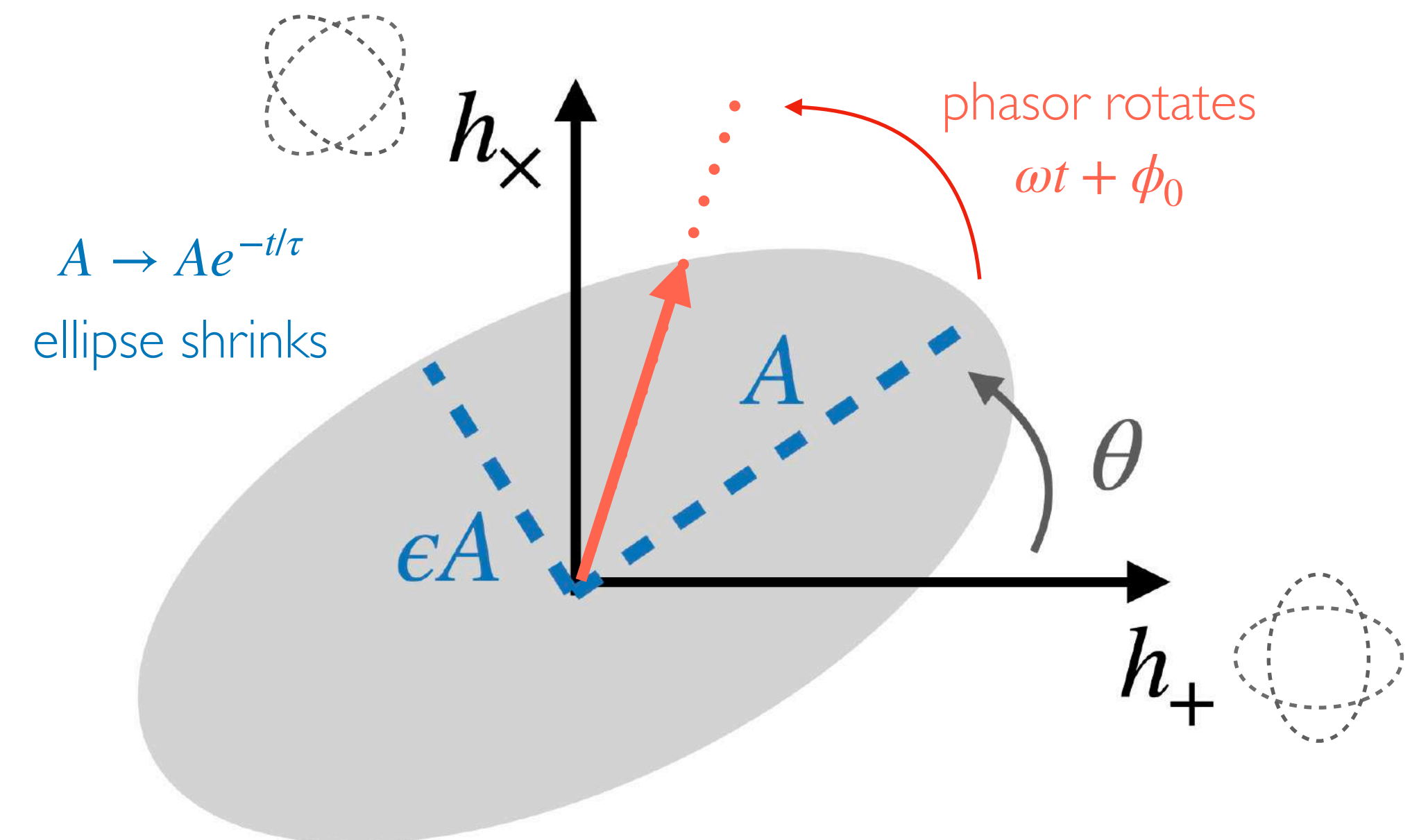


all solutions to the Teukolsky equation

ringdown

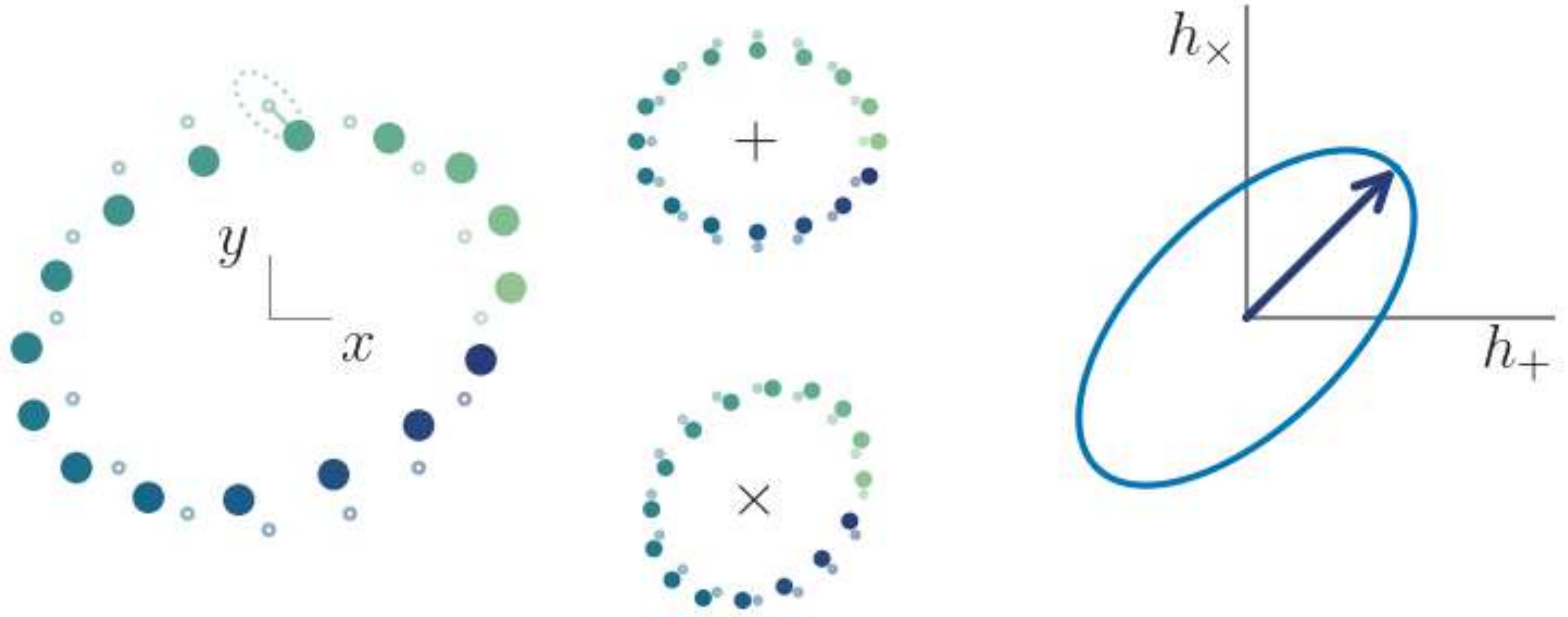
$$h \equiv h_+ - ih_\times \quad , \quad j \equiv \{\ell, |m|, n\}$$

$$h = \sum_j \frac{1}{2} A_j e^{-t/\tau_j} \left[(1 + \epsilon_j) e^{-i(\omega_j t - \phi_{+j})} + (1 - \epsilon_j) e^{i(\omega_j t + \phi_{-j})} \right]$$



each QNM appears as an **elliptically-polarized** mode

ellipse would shrink over time due to QNM decay



test masses

linear polarizations

polarization phasor

ringdown

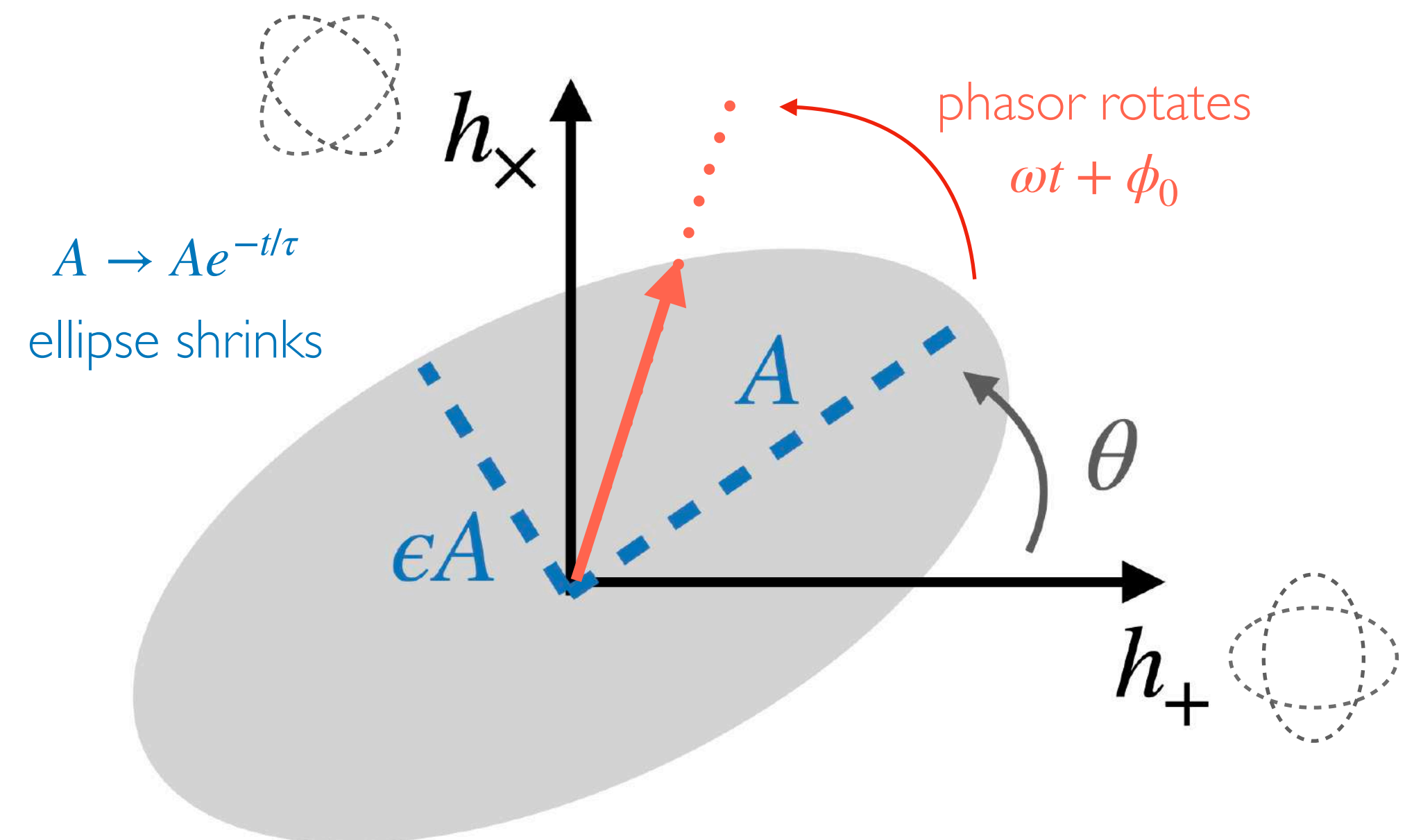
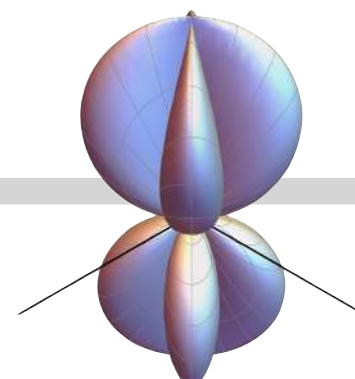
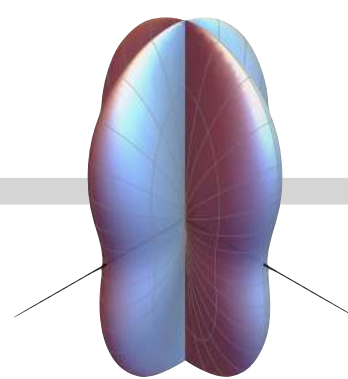
caveat :
+ other stuff

$$h \equiv h_+ - ih_\times \quad , \quad j \equiv \{\ell, |m|, n\}$$

$$h = \sum_j \frac{1}{2} A_j e^{-t/\tau_j} \left[(1 + \epsilon_j) e^{-i(\omega_j t - \phi_{+j})} + (1 - \epsilon_j) e^{i(\omega_j t + \phi_{-j})} \right]$$

strain at I^{th} detector

$$h_I(t) = F_+^I(\alpha, \delta) h_+(t) + F_\times^I(\alpha, \delta) h_\times(t)$$

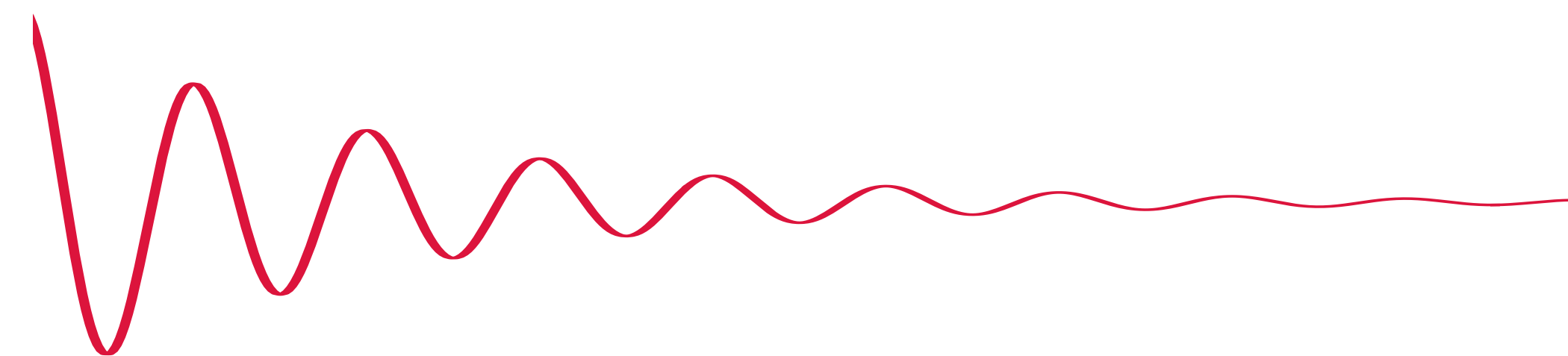


black-hole spectroscopy

Kerr spectrum depends only on mass and spin

measure BH parameters from GW signal

extract frequency and damping of multiple modes to check **consistency with Kerr**

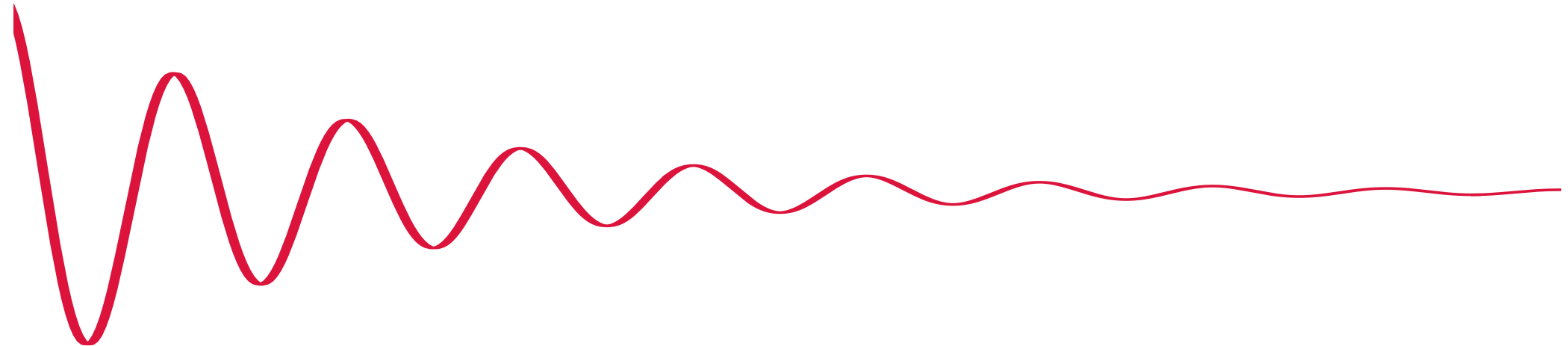
$$h = \sum_{\ell mn} C_{\ell mn} e^{-i\tilde{\omega}_{\ell mn} t}$$


black-hole spectroscopy

amplitudes and phases encode initial conditions

make inferences about **pre-merger phase**

identify traces of misaligned spins, high mass ratios
or eccentricity, all **astrophysical important**

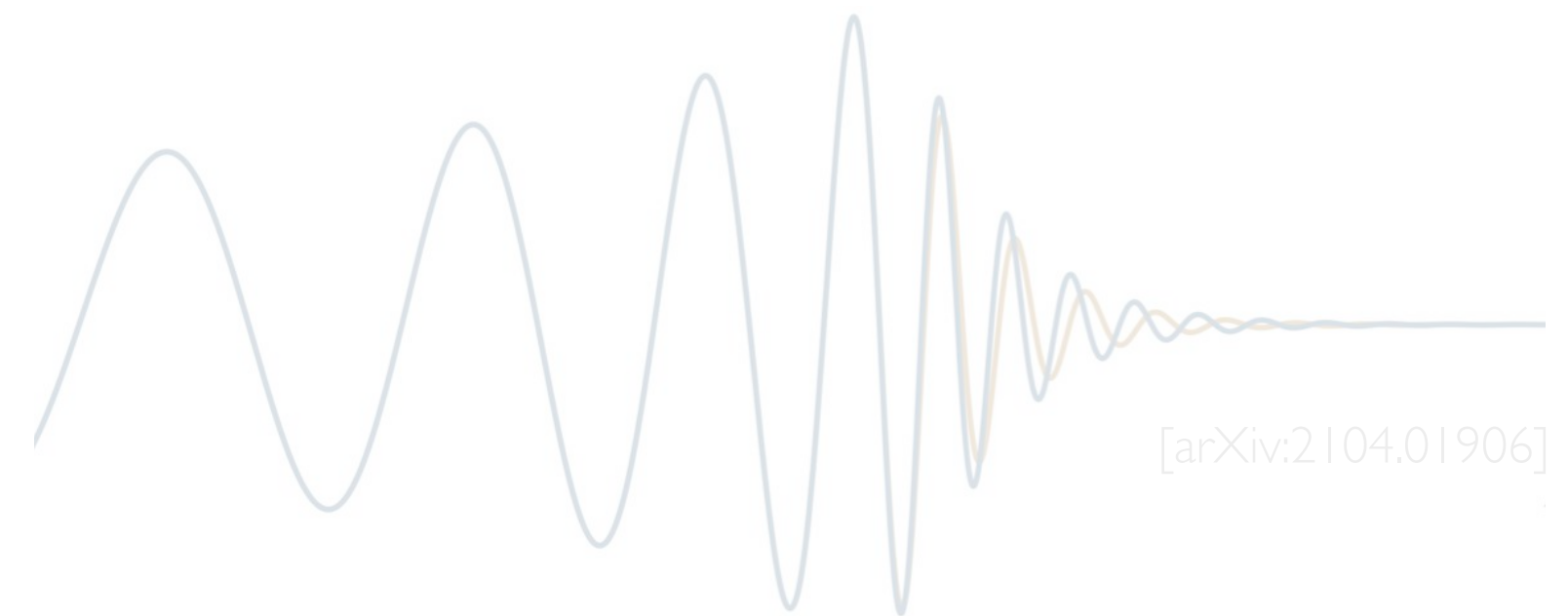
$$h = \sum_{\ell mn} C_{\ell mn} e^{-i\tilde{\omega}_{\ell mn} t}$$


strategies

enhanced IMR analysis

construct IMR model with
additional degrees of freedom

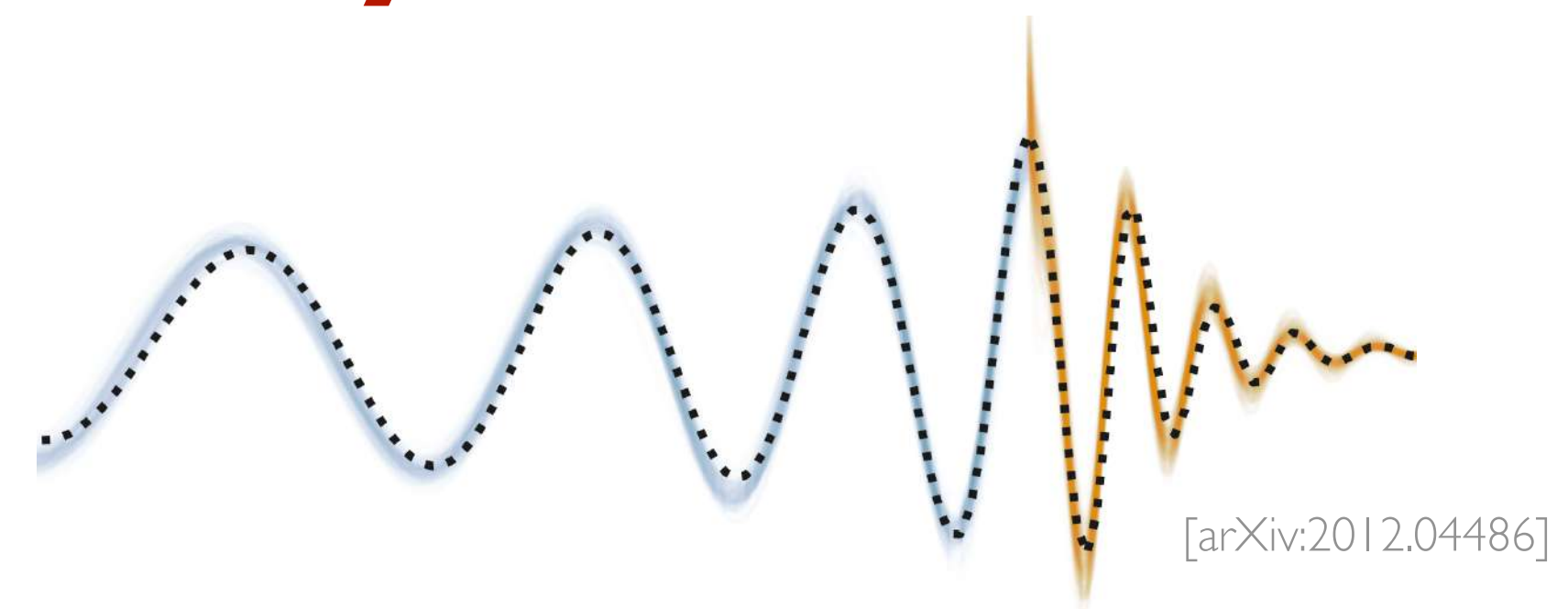
Brito+2018 [arXiv:1805.00293] Ghosh+2021 [arXiv:2104.01906]



ringdown-only analysis

isolate ringdown data and fit
damped sinusoids

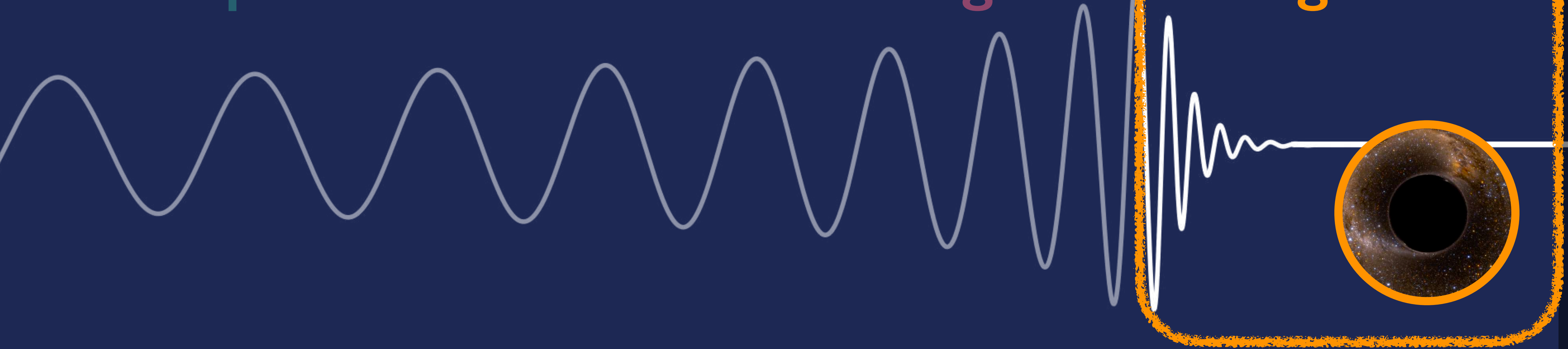
Isi+2019 [arXiv:1905.00869] Isi+2021 [arXiv:2107.05609]
Carullo+2019 [arXiv:1902.07527]



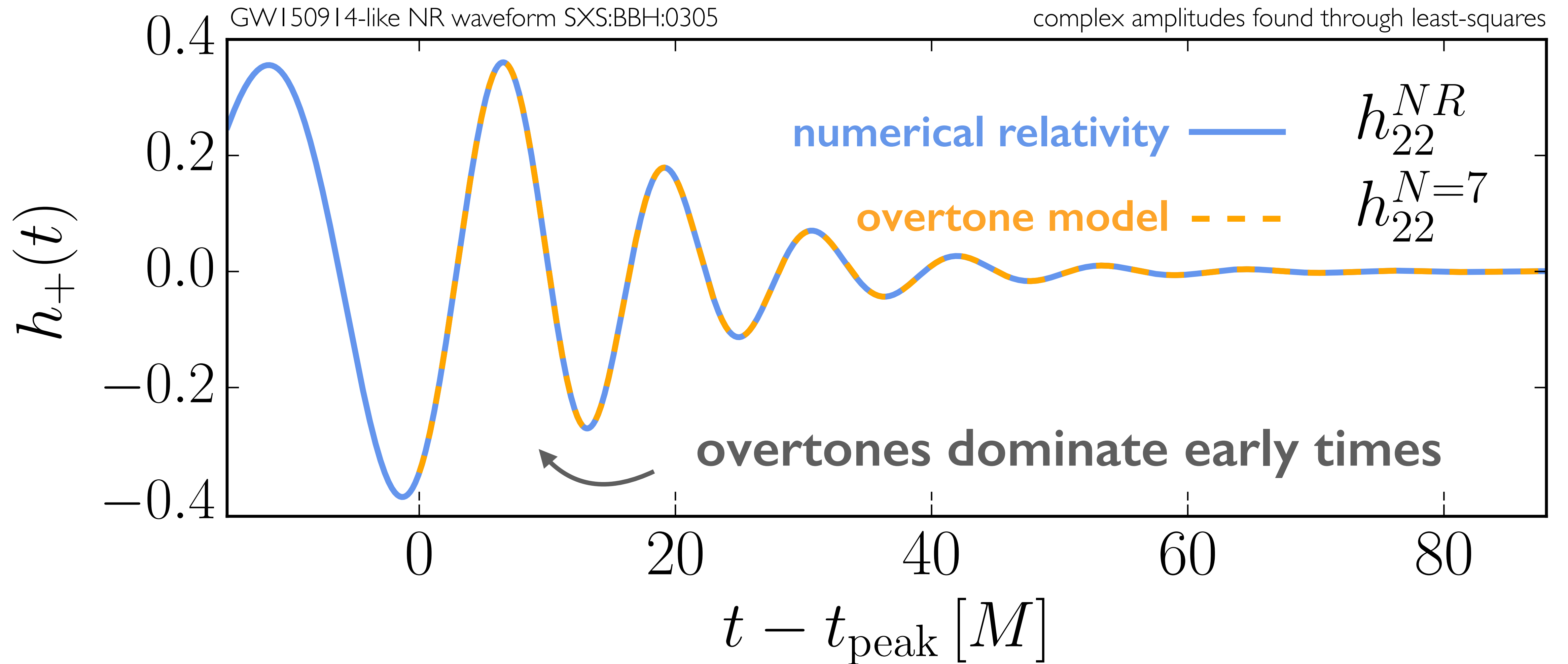
inspiral

merger

ringdown?



ringdown starts at the peak?



a (biased) history

a subjective summary of ringdown measurements:

post-2019 Giesler+ and Isi+ tantalizing...
but quickly become **controversial**

Theory surely the spacetime is nonlinear at the peak! Overtones can be overfitting NR.

Observation overtones are difficult to fit and others don't seem able to find them in GW150914.



inspired by Vishal Baibhav's meme at 2022
GWPAW talk

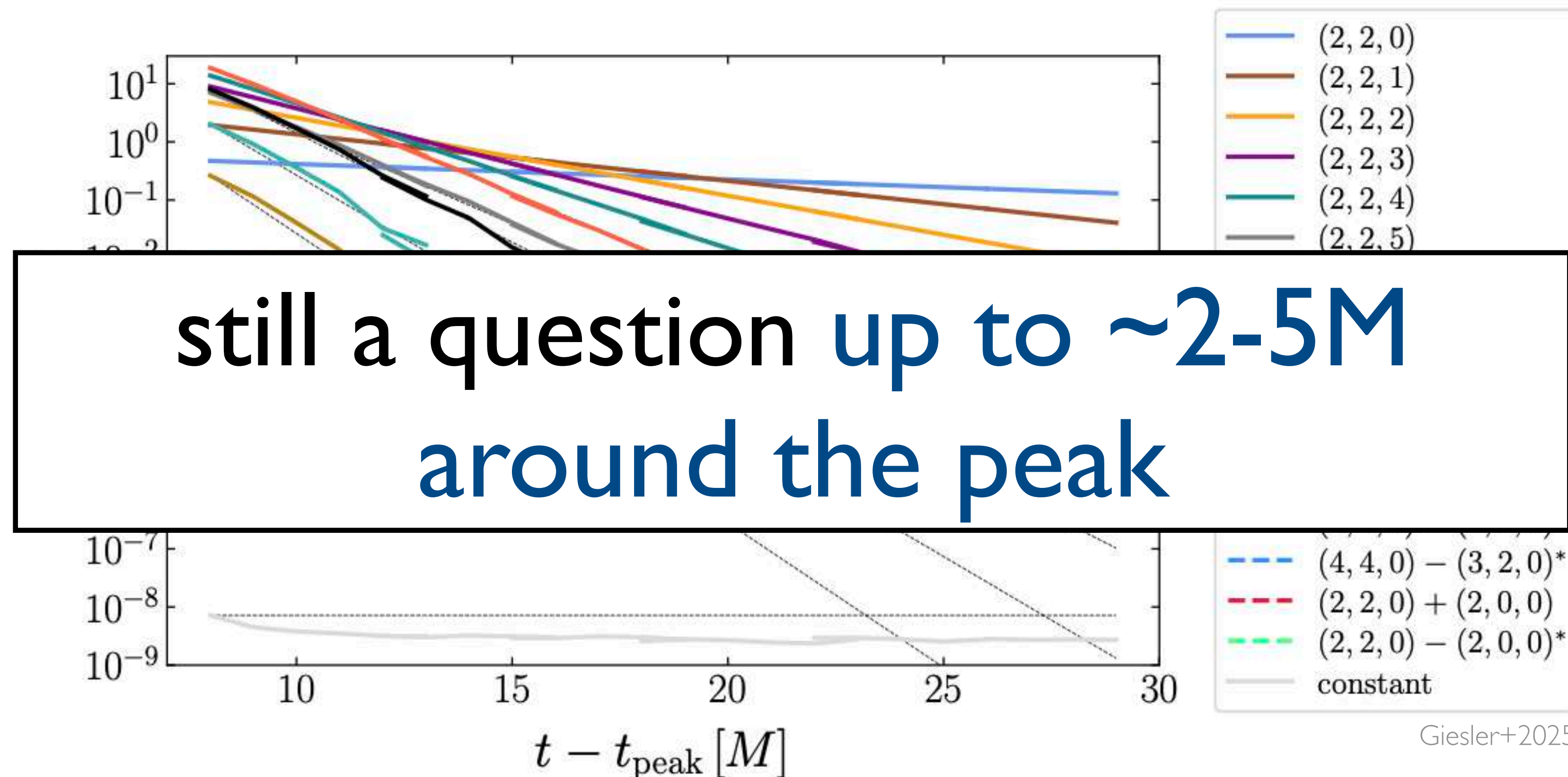
fully understood
Isi & Farr (PRL 2023)
[arXiv:2310.13869]

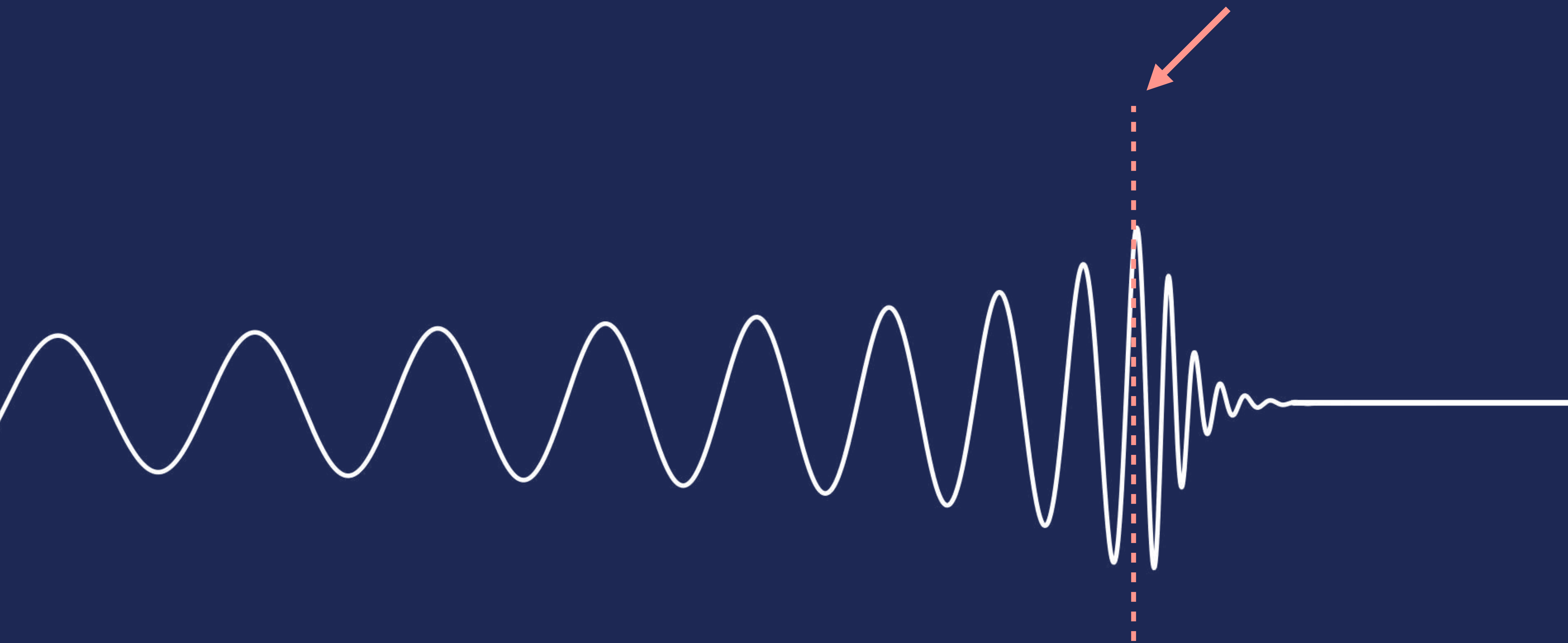
a (biased) history

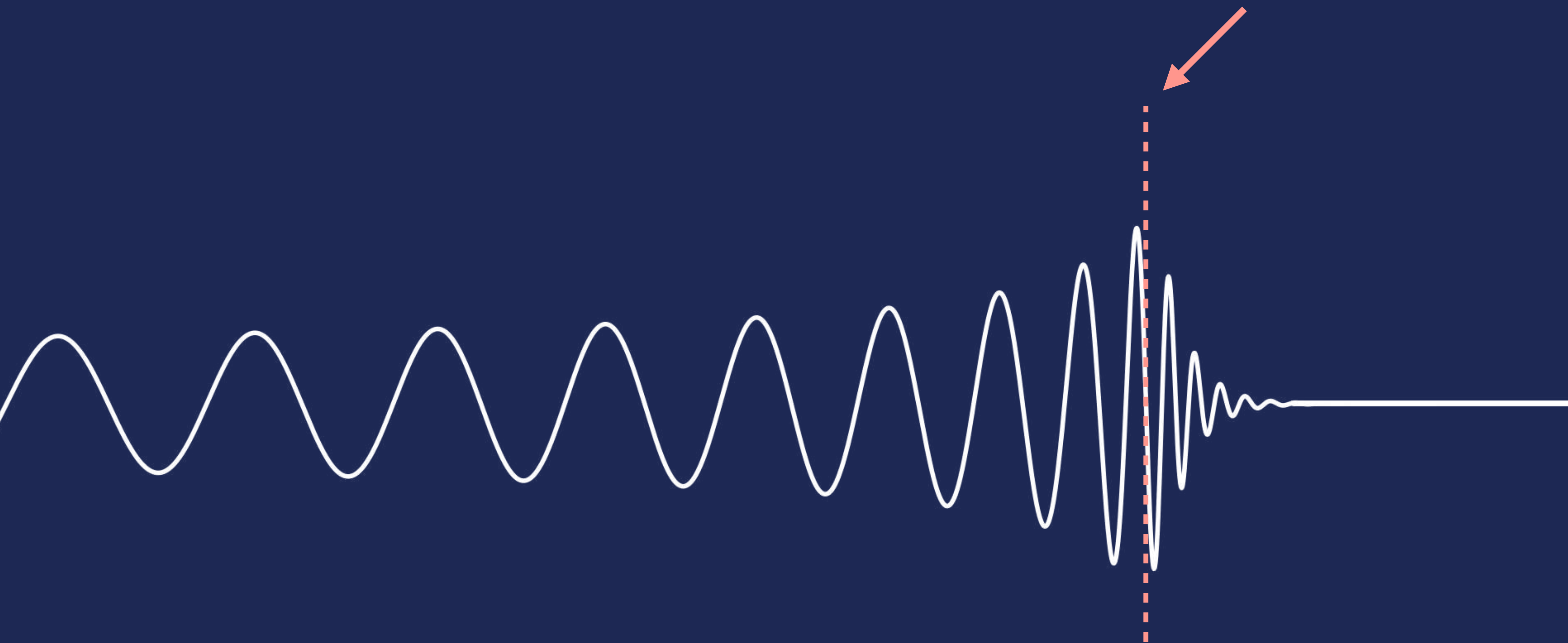
Theory surely the spacetime is nonlinear at the peak!

Overtone can be overfitting NR **[still debatable]**

see, e.g., Giesler+2025 and Mitman+2025; also *lots* of work by Baibhav+, Cheung+, Finch+, Dyer+, ... review [arXiv:2505.23895]







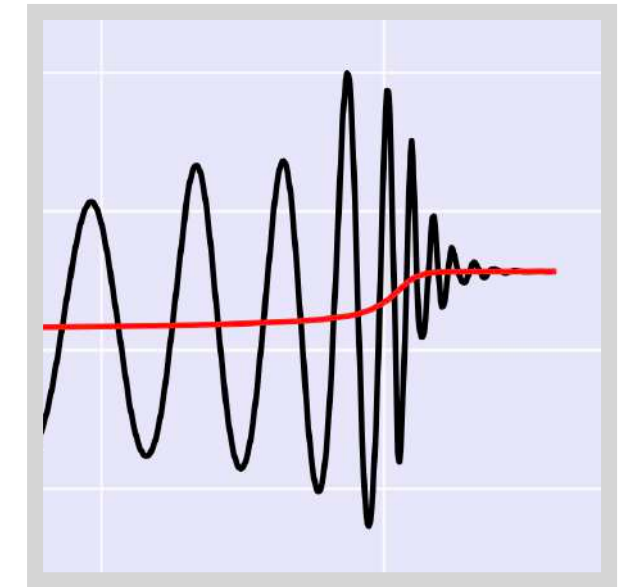
post-peak signal looks
really close to linear
even if not exactly so



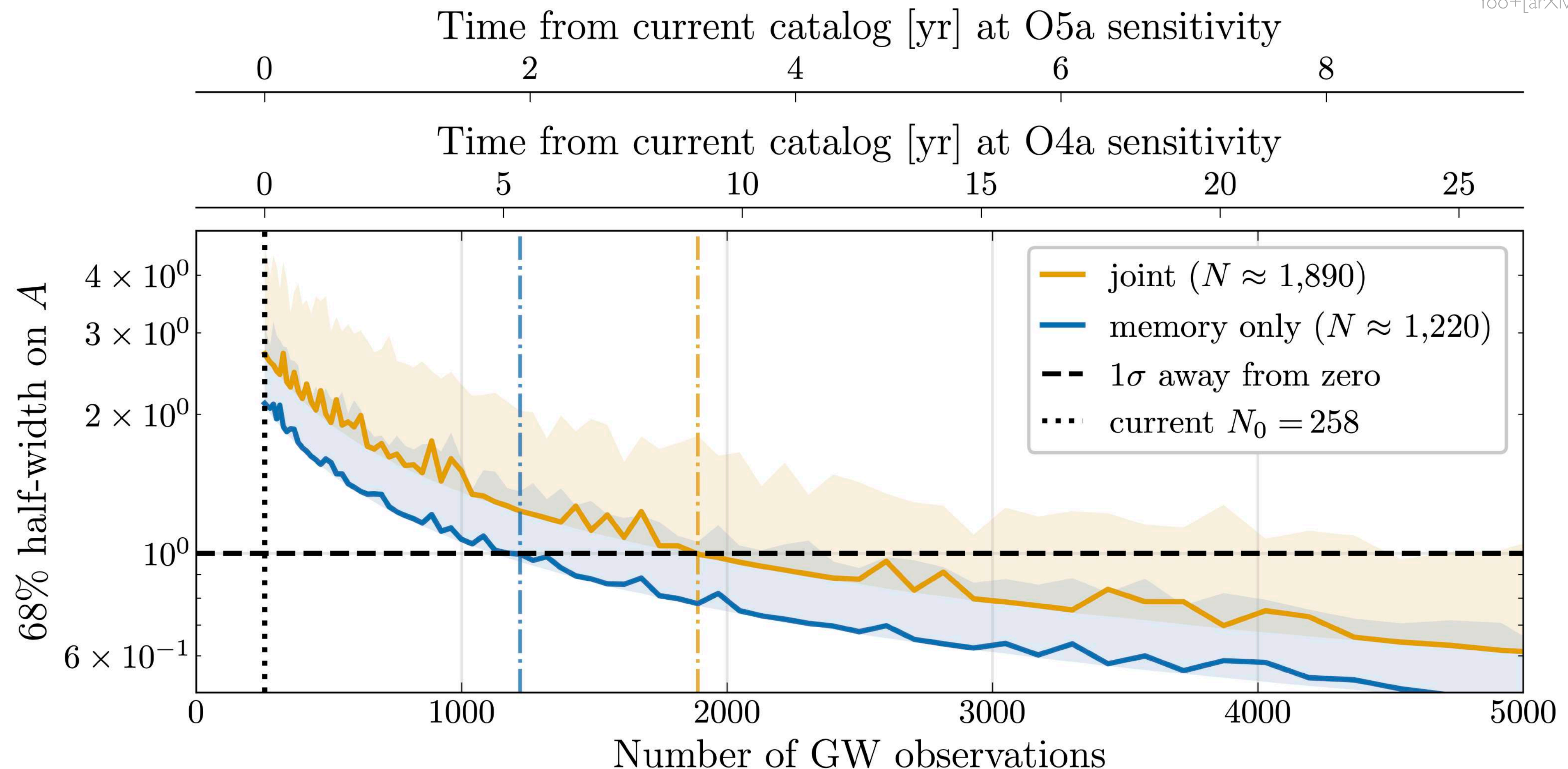
for *real* GW observations, all we want is
systematic error < statistical uncertainty
we don't need a perfect model!

memory measurement

K. Mitman, M. Isi, W. Farr [arXiv:2605.27500] → **achievable by O5!**



Yoo+[arXiv:2306.03148]



observations

two types of systems

lighter



GW150914-like
 $M_f \sim 70 M_\odot$

heavier



GW190521-like
 $M_f \sim 200 M_\odot$

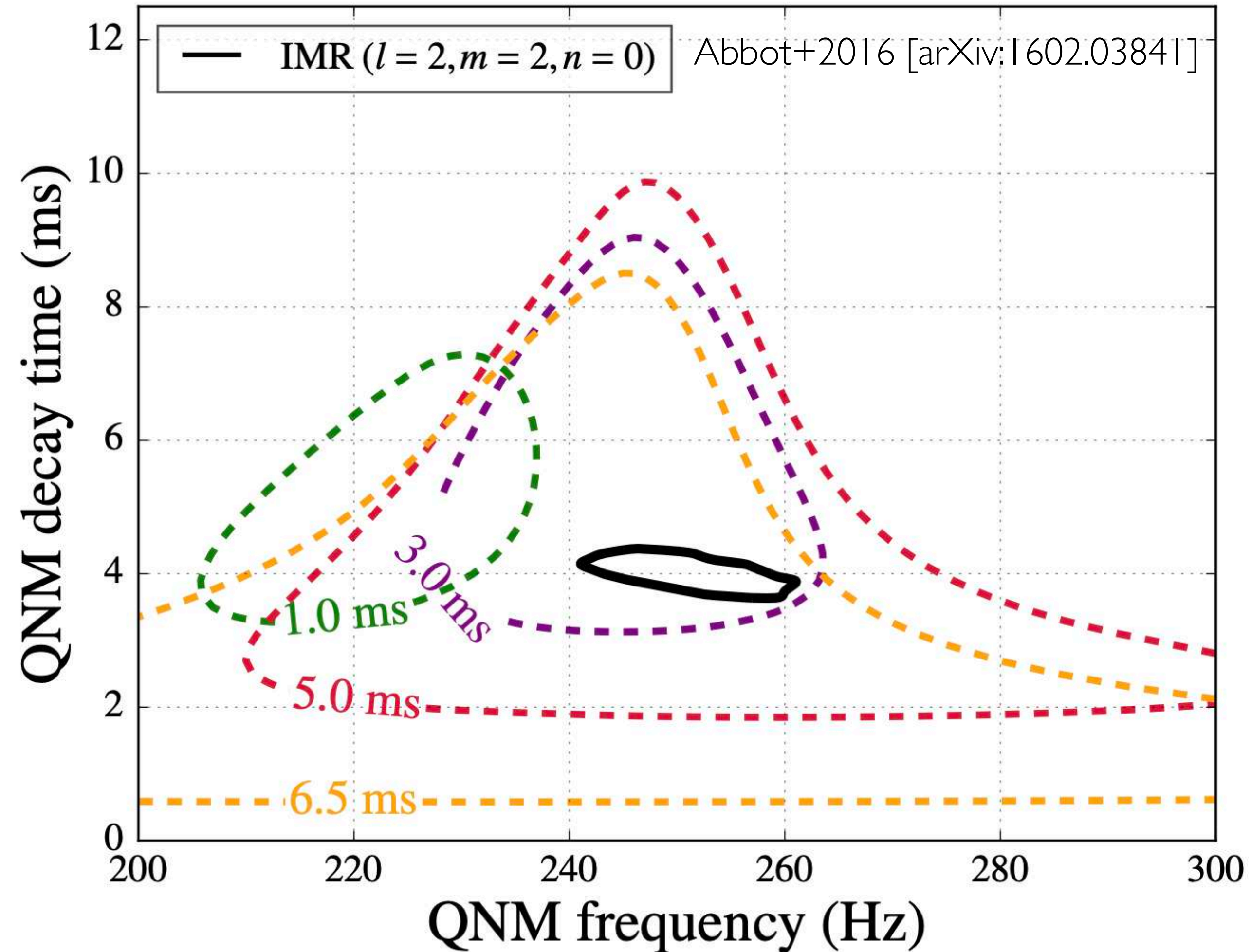
lighter systems

GW150914

GW250114



LIGO-Virgo analysis in 2016



GW150914 at the peak



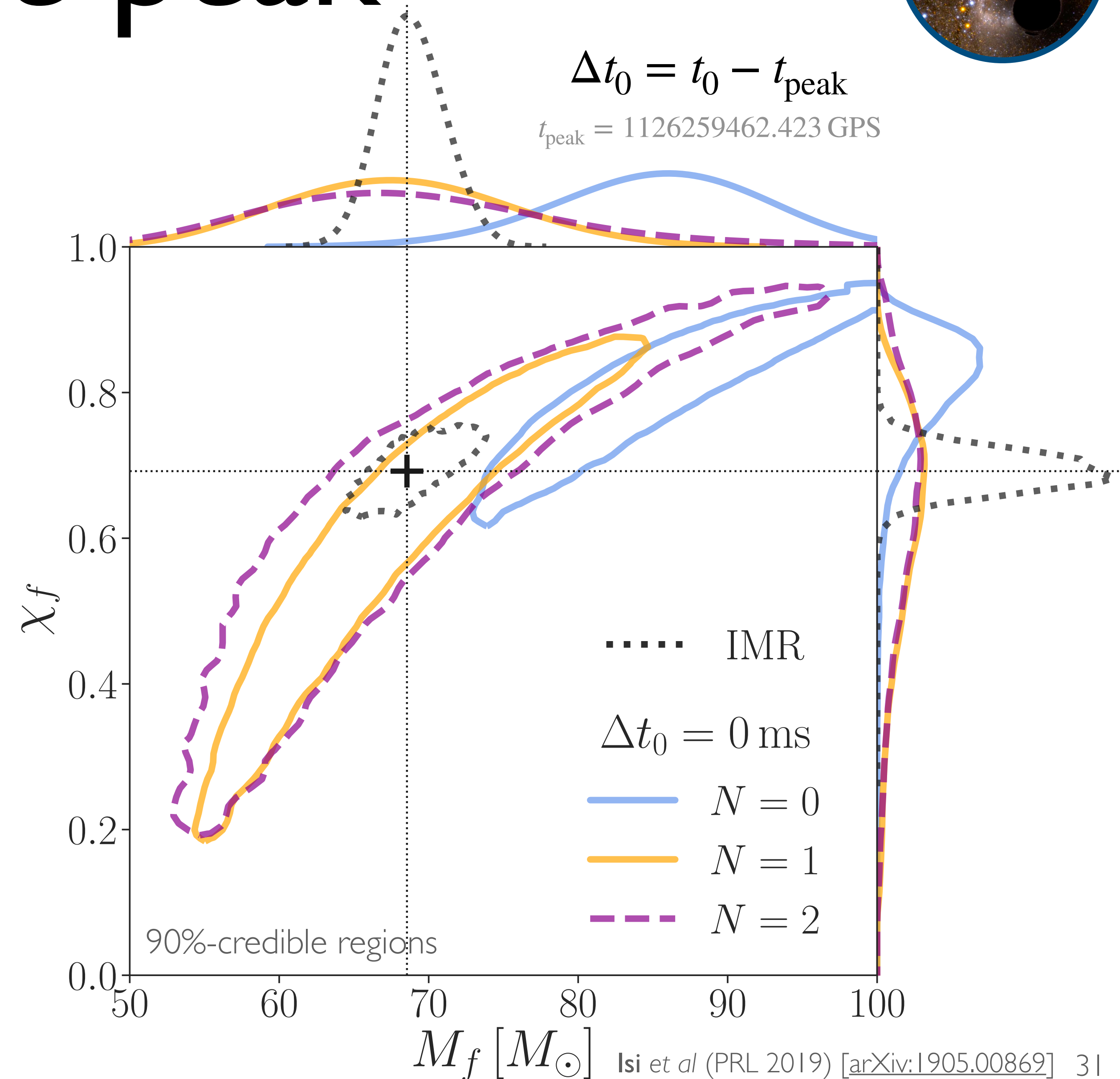
assuming damped sinusoids

we identify at least **two ringdown modes** in GW150914

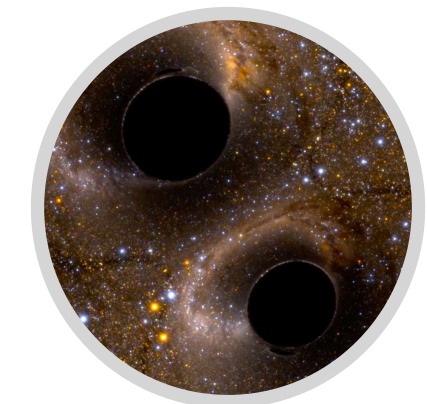
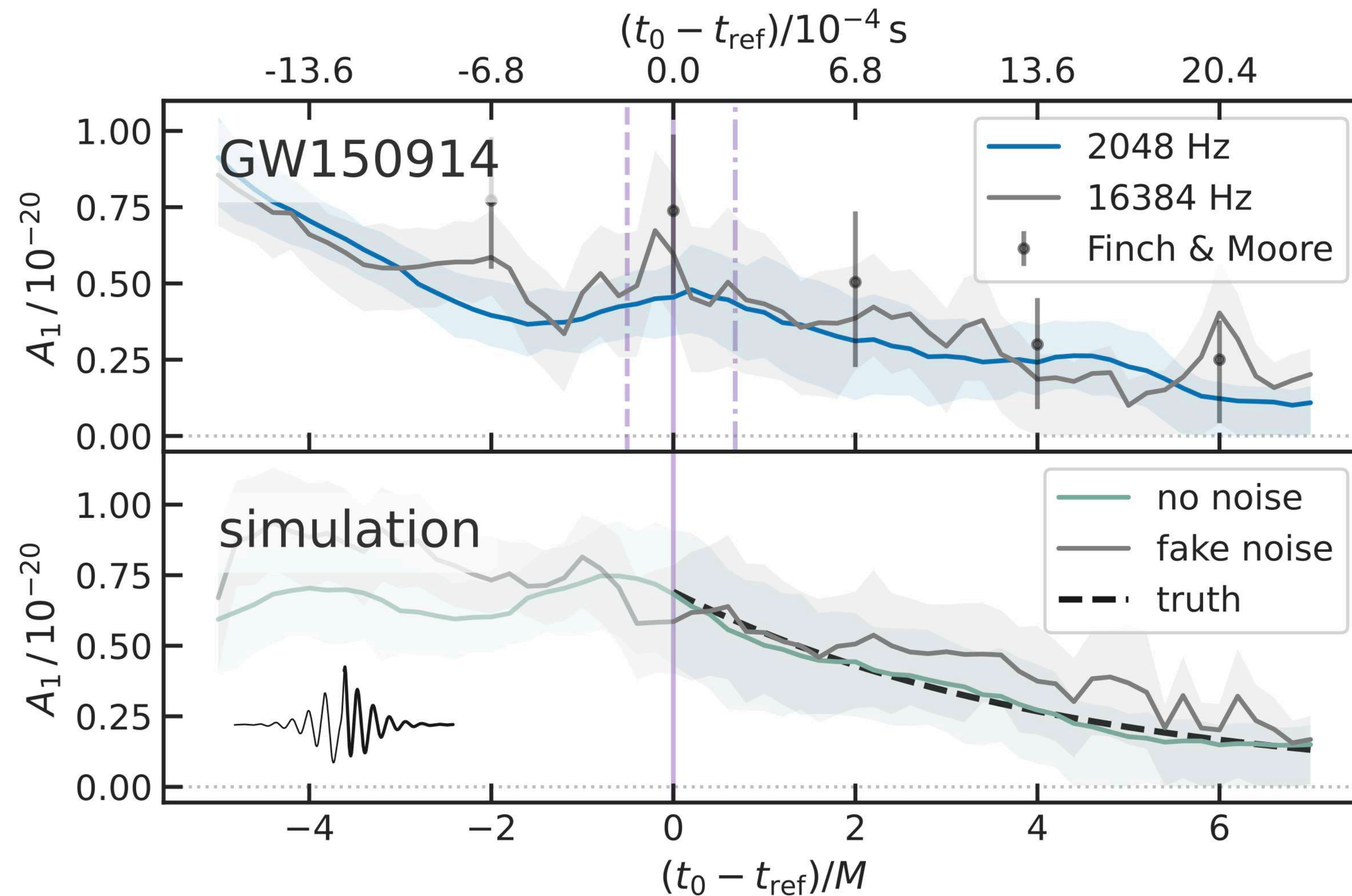
that is the least-damped mode plus **at least one overtone** ($N=1$)

with $N=1$, can **measure remnant mass and spin** at the peak

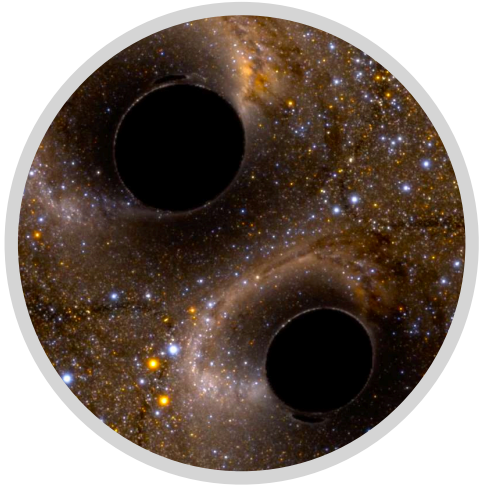
result in agreement with measurement from full waveform



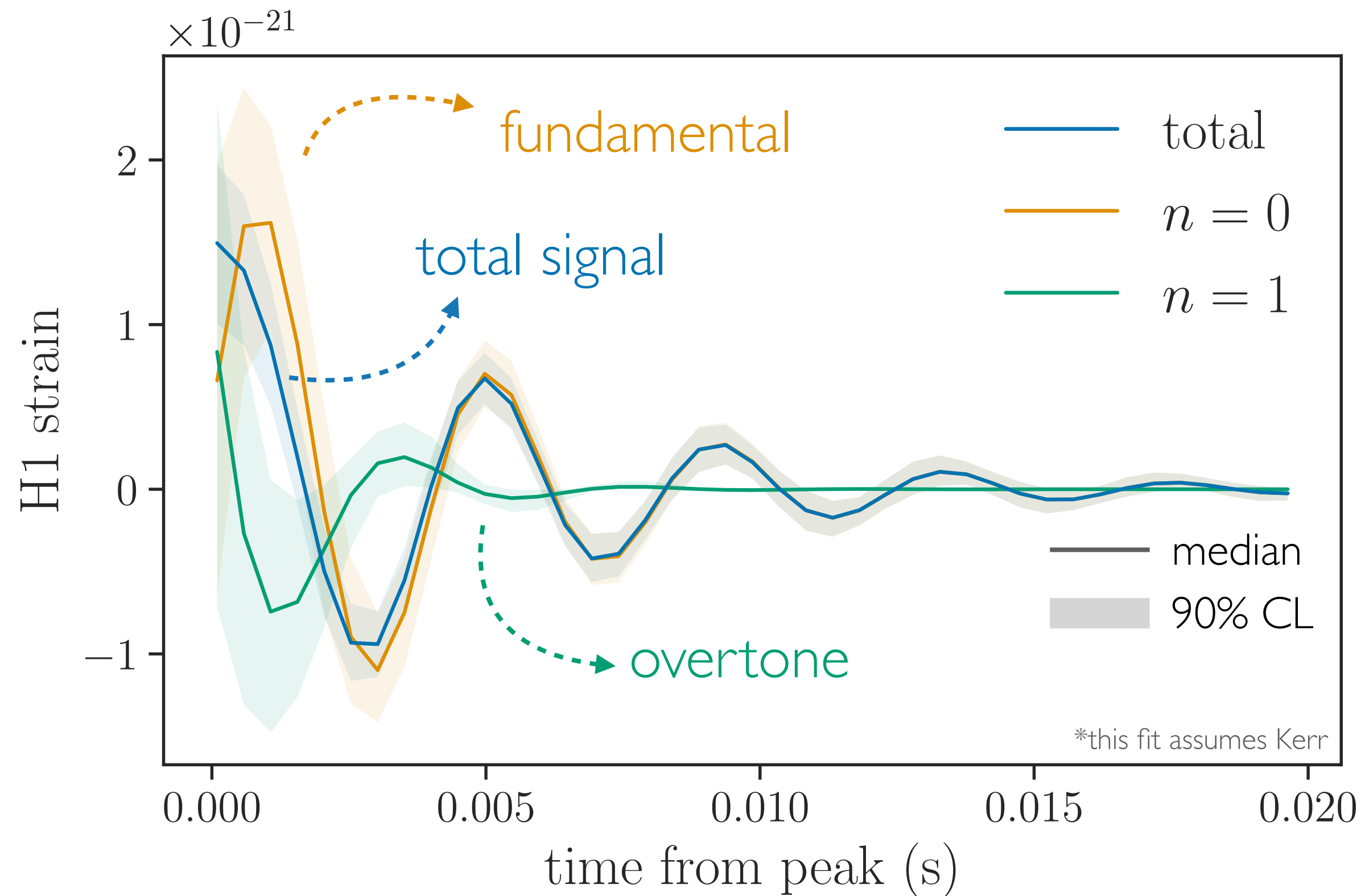
overtone amplitude over time



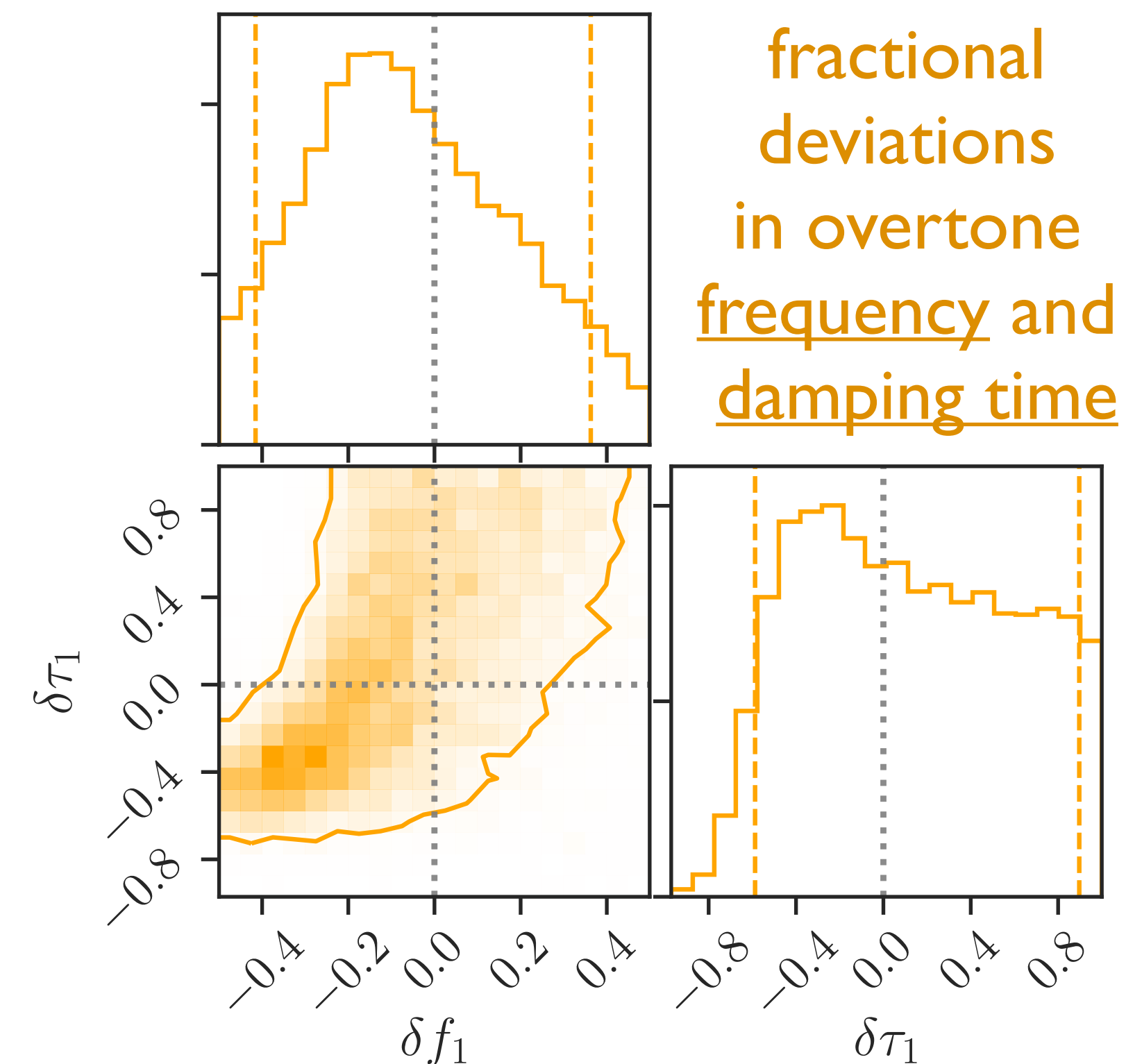
GW150914



spectroscopic analysis of fundamental + first overtone



reconstruct fundamental and
overtone of 22 mode



agreement with Kerr
~ 20% at 1σ

10 Years Later: LIGO Hears Loud and Clear

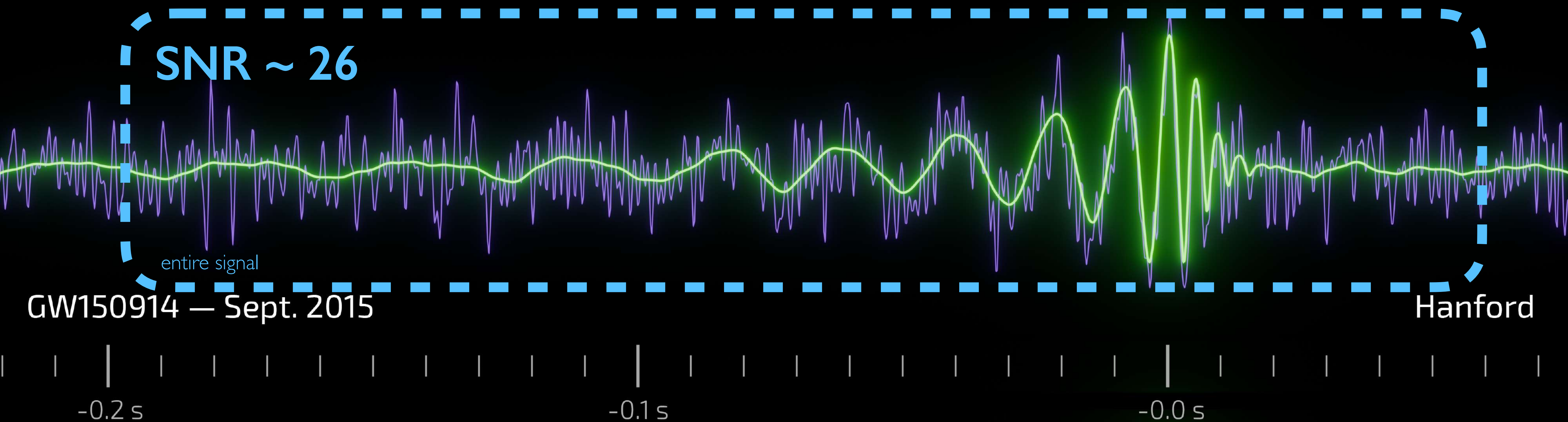
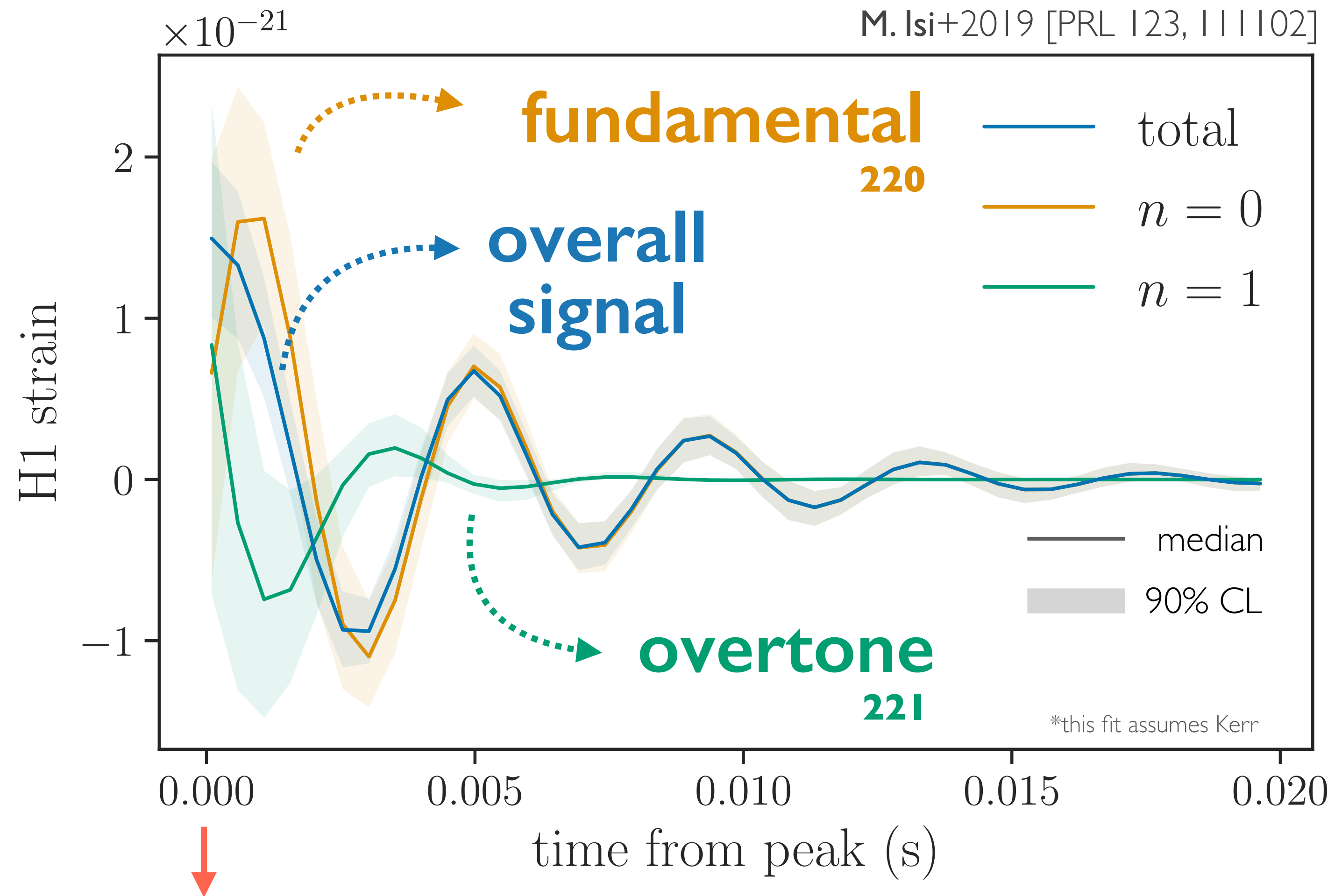
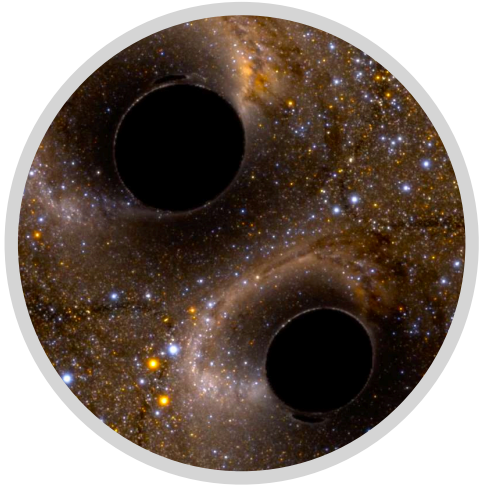


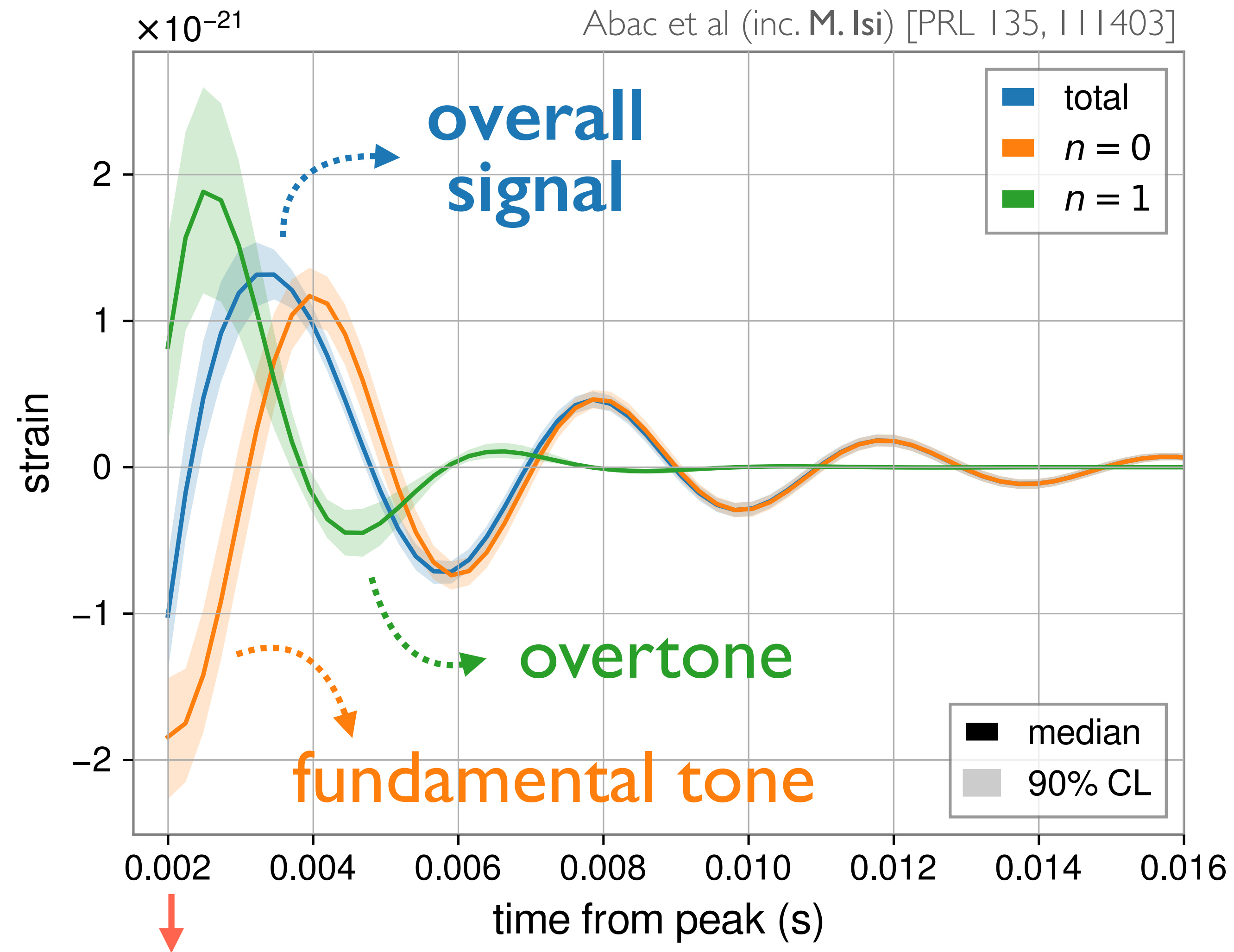
Image credit: LIGO/J. Tisso (GSSI)/R. Hurt (Caltech-IPAC)

GW150914



OM after peak

GW250114

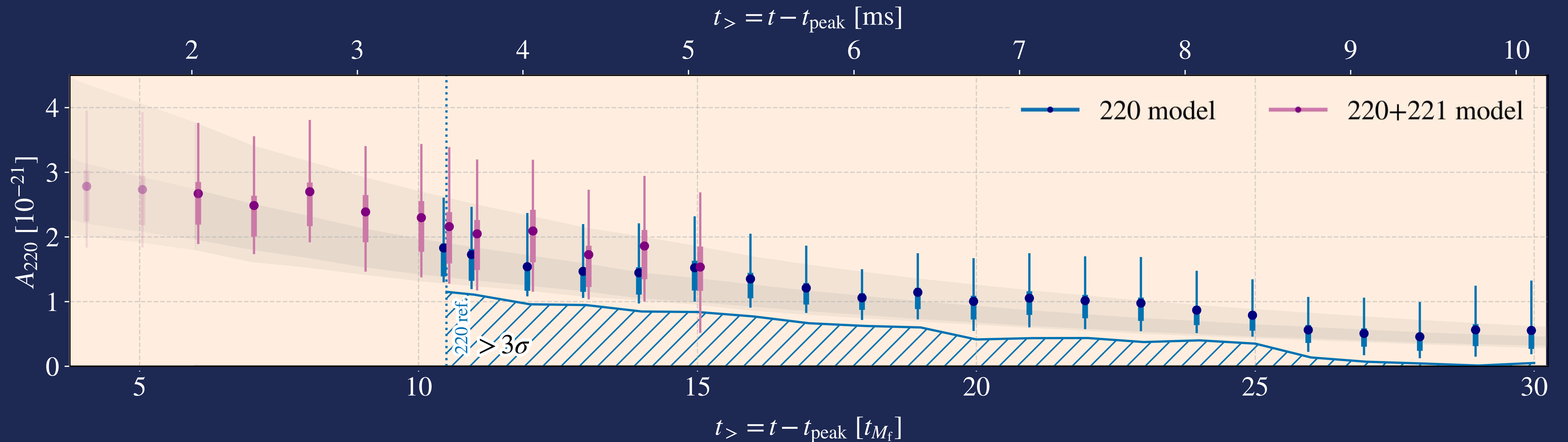


6M after peak

GW250114 ringdown



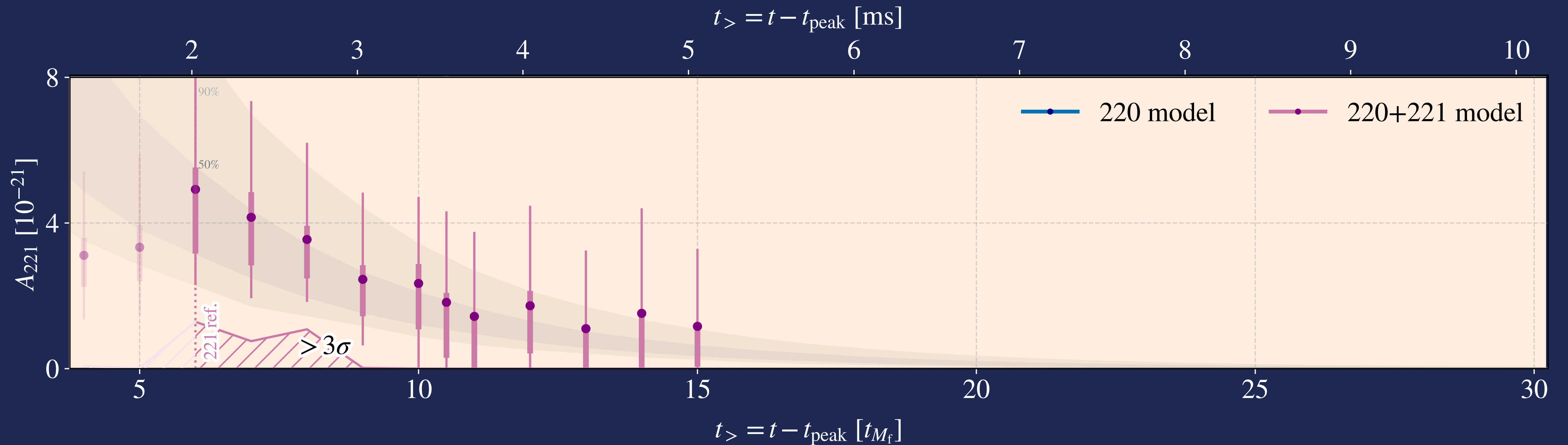
fundamental amplitude



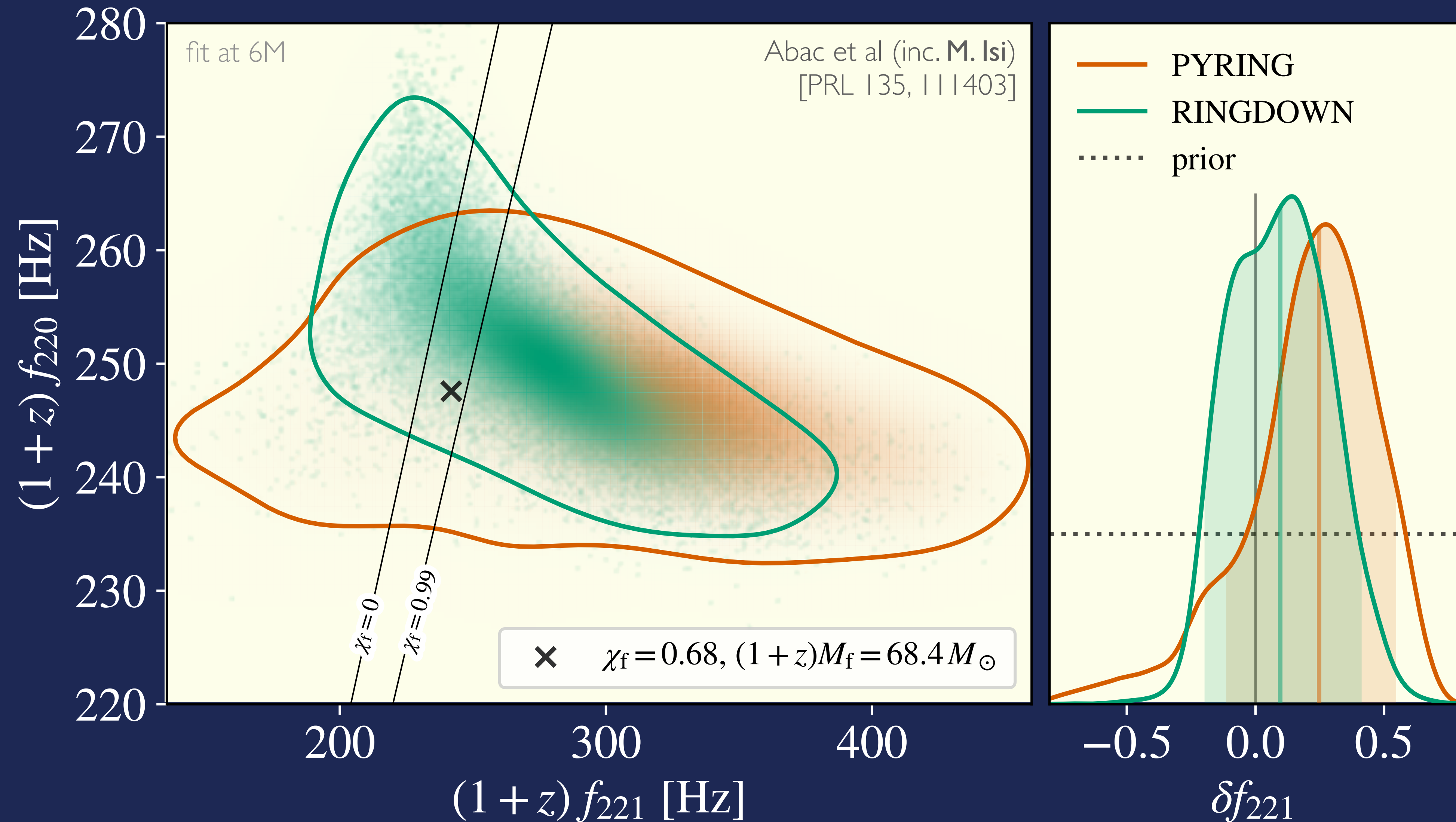
GW250114 ringdown



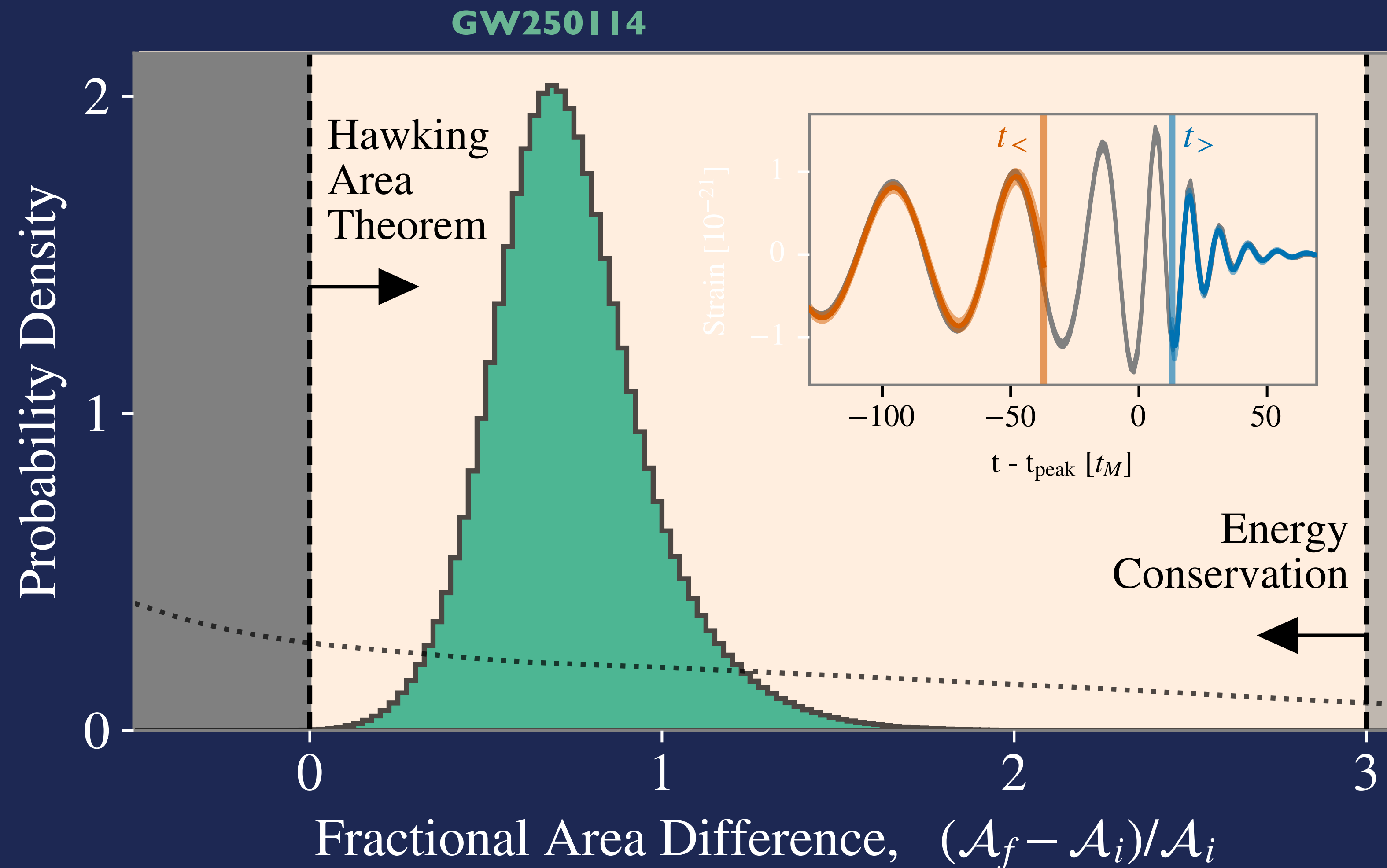
overtone amplitude



test of **Kerr nature** of the remnant black hole



GW250114 area law



final state is
consistent
with initial
state

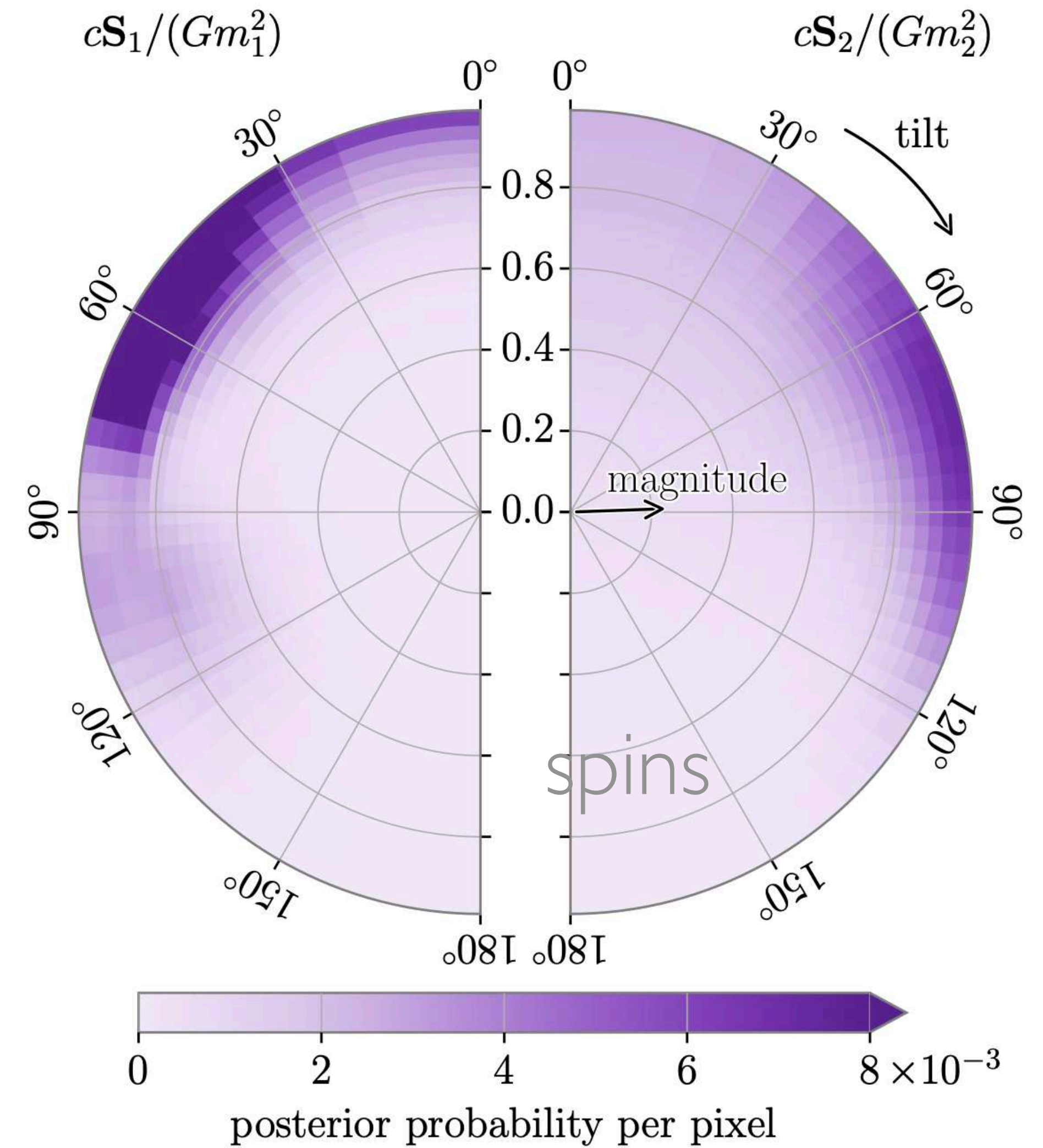
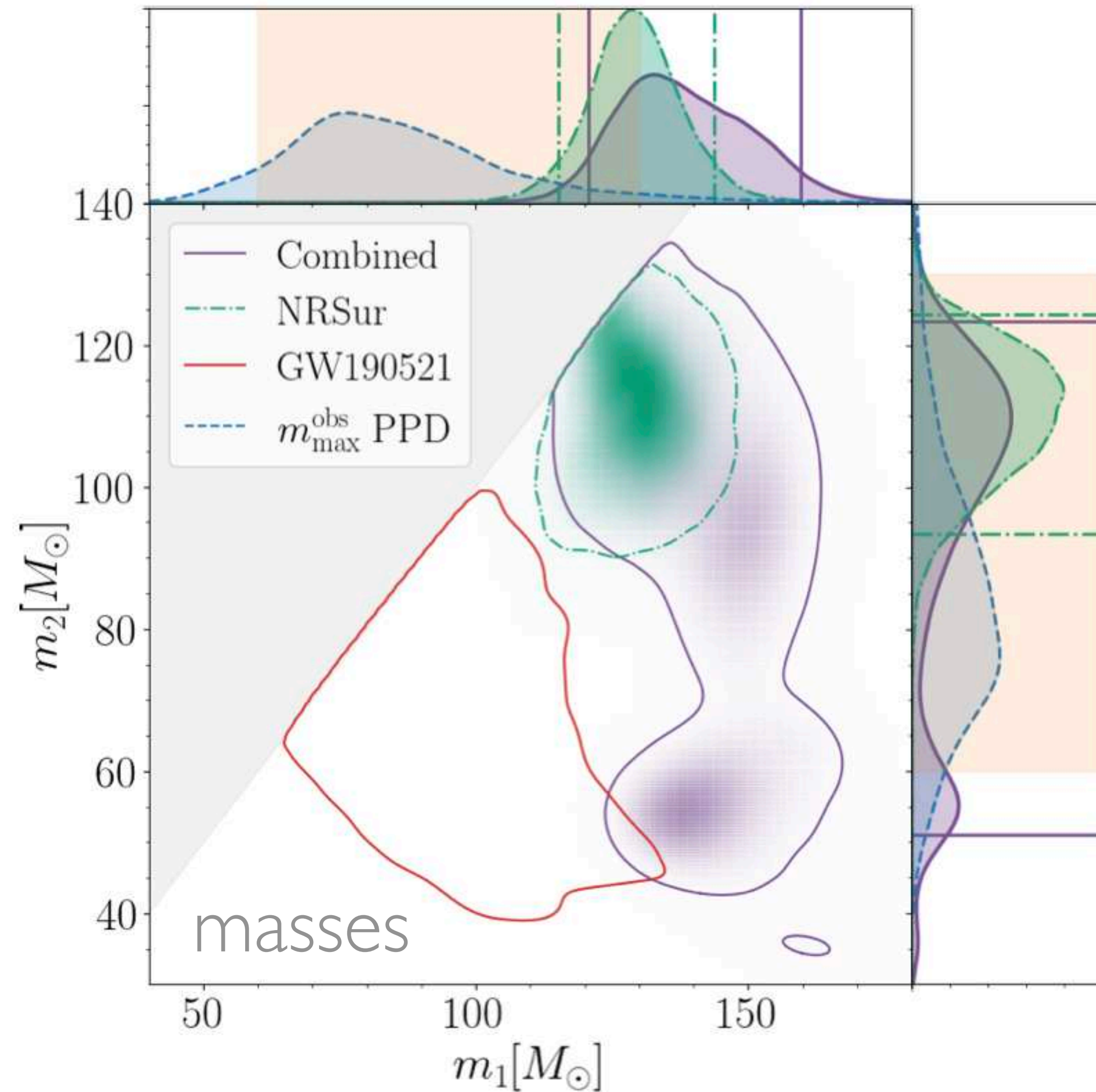
heavier systems

GW190521

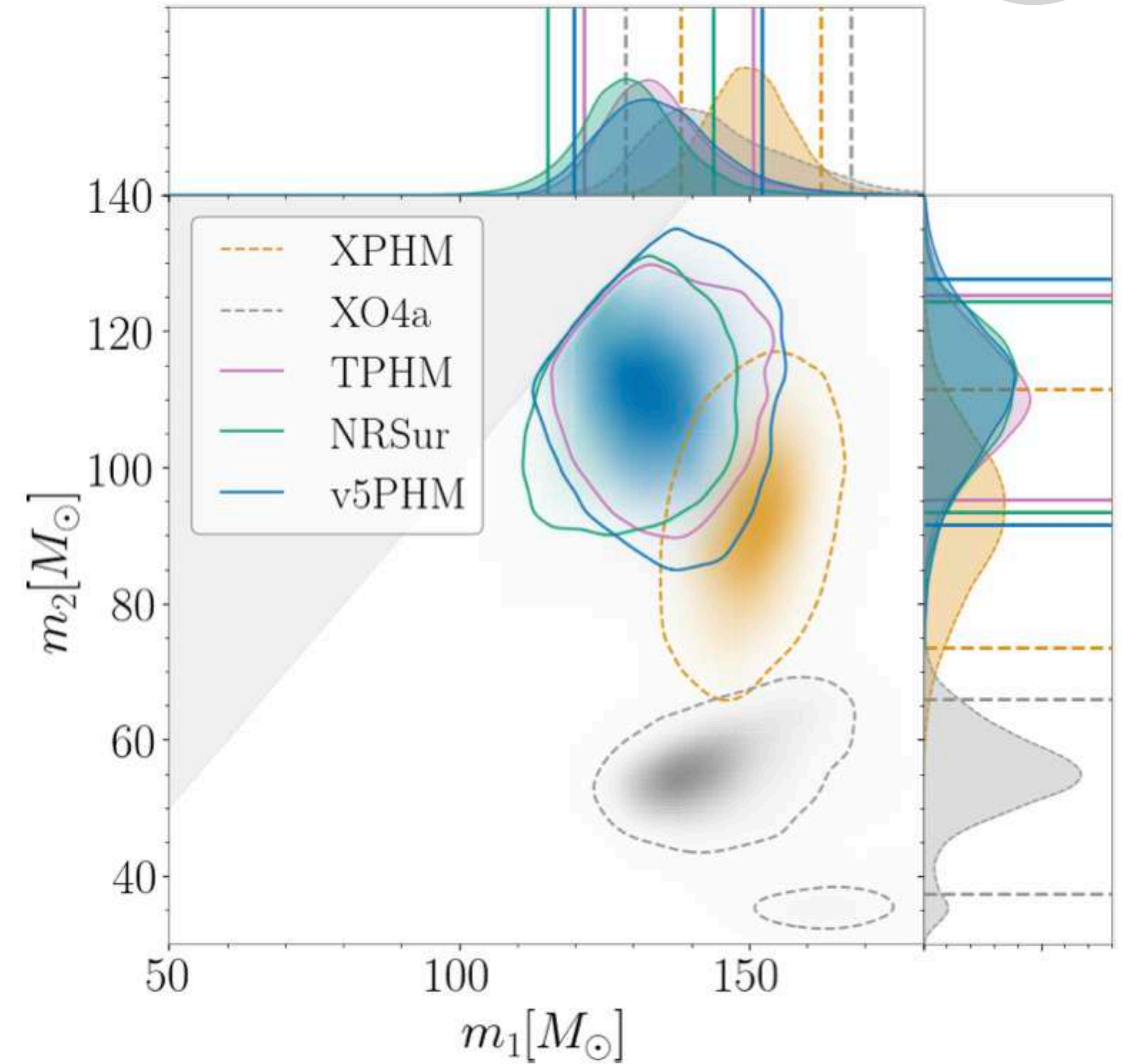
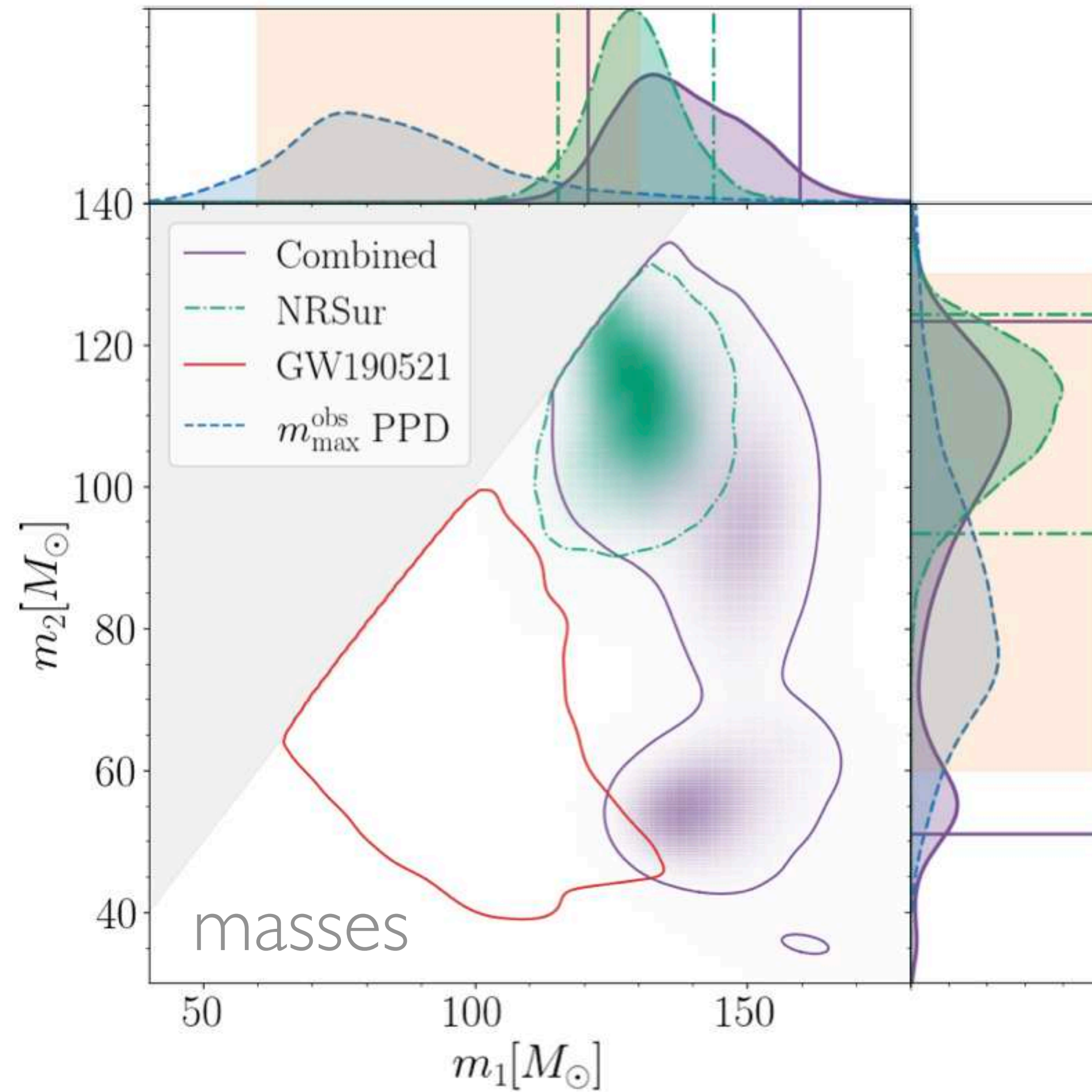
GW231123



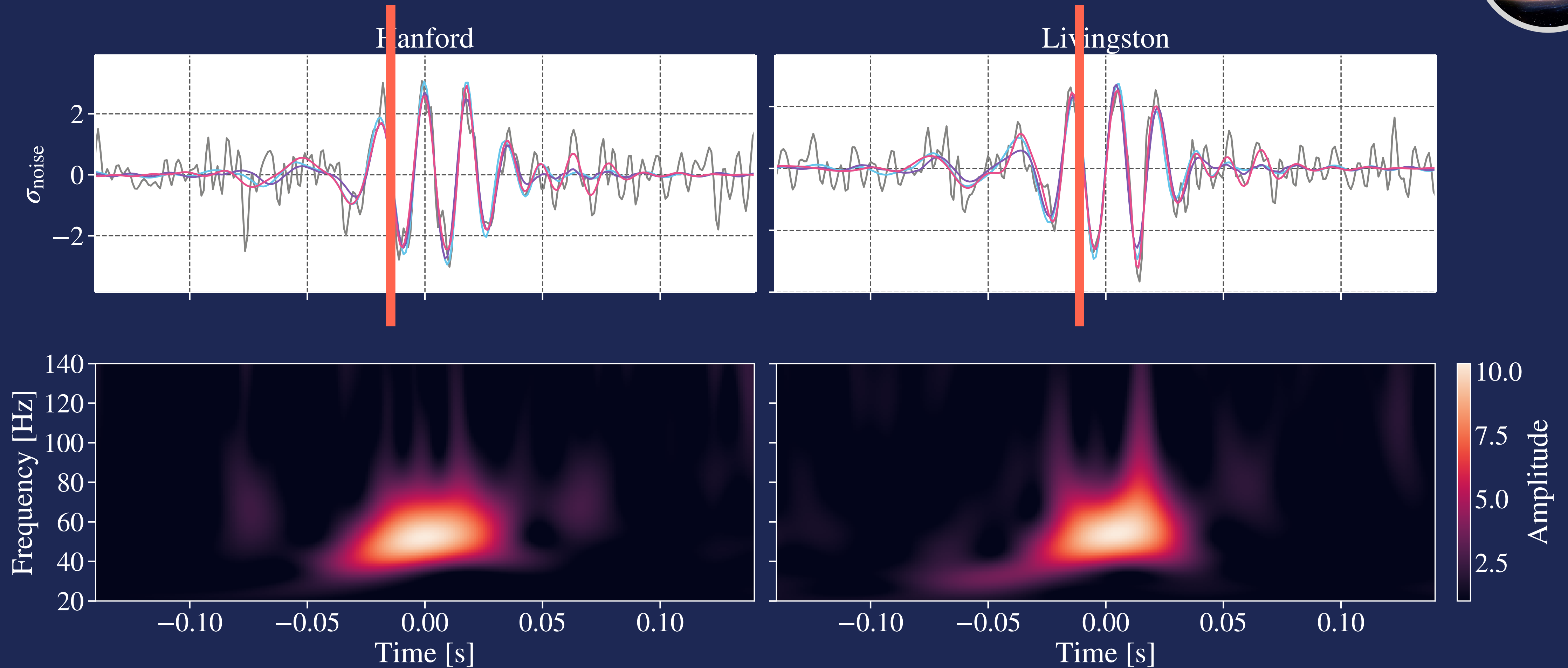
GW231123



GW231123



GW231123



— Whitened data ■ BBH Template Reconstruction (Bilby) ■ Wavelet Reconstruction (BayesWave) — cWB Reconstruction



heavy black holes

merger

dominated

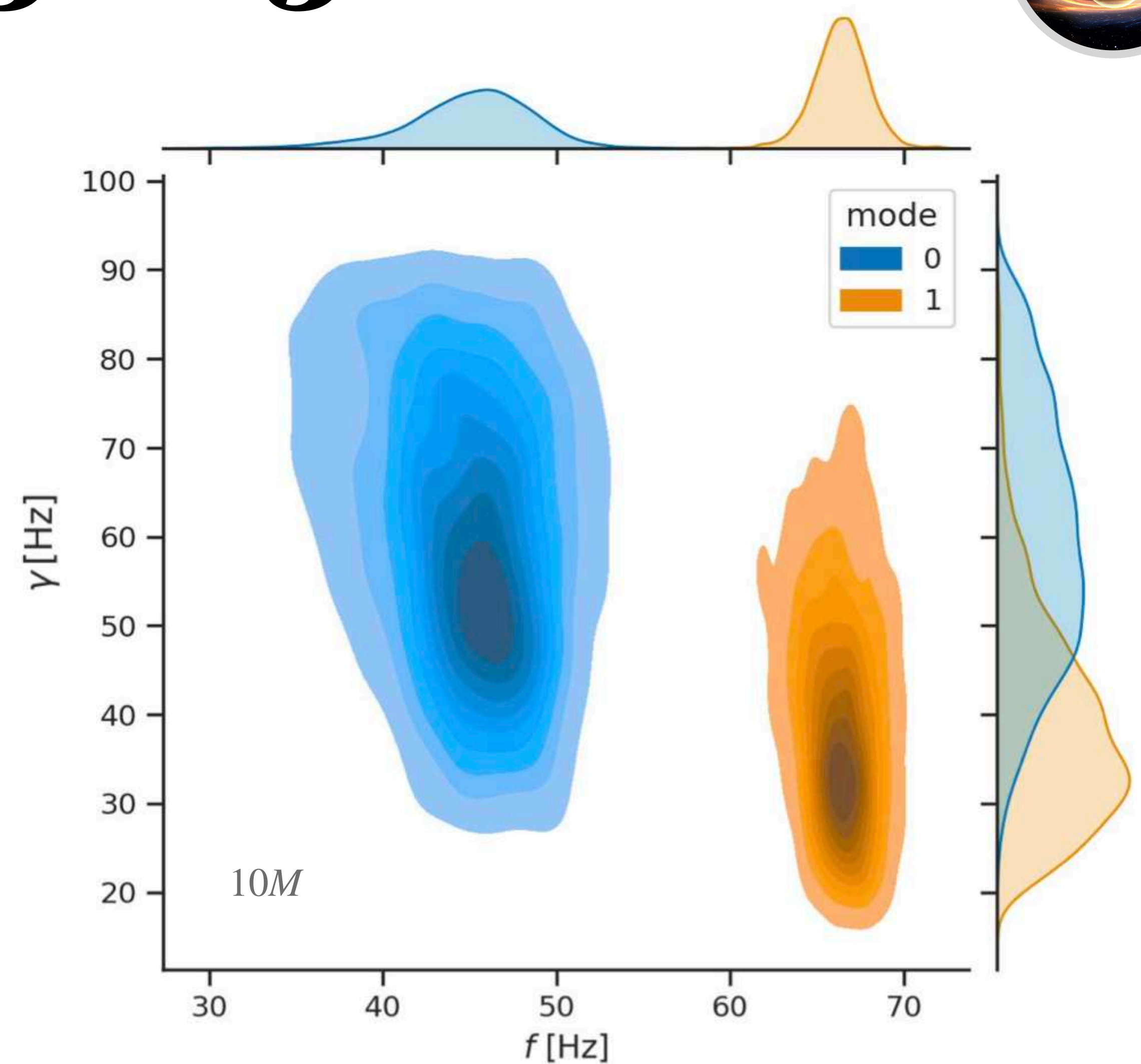
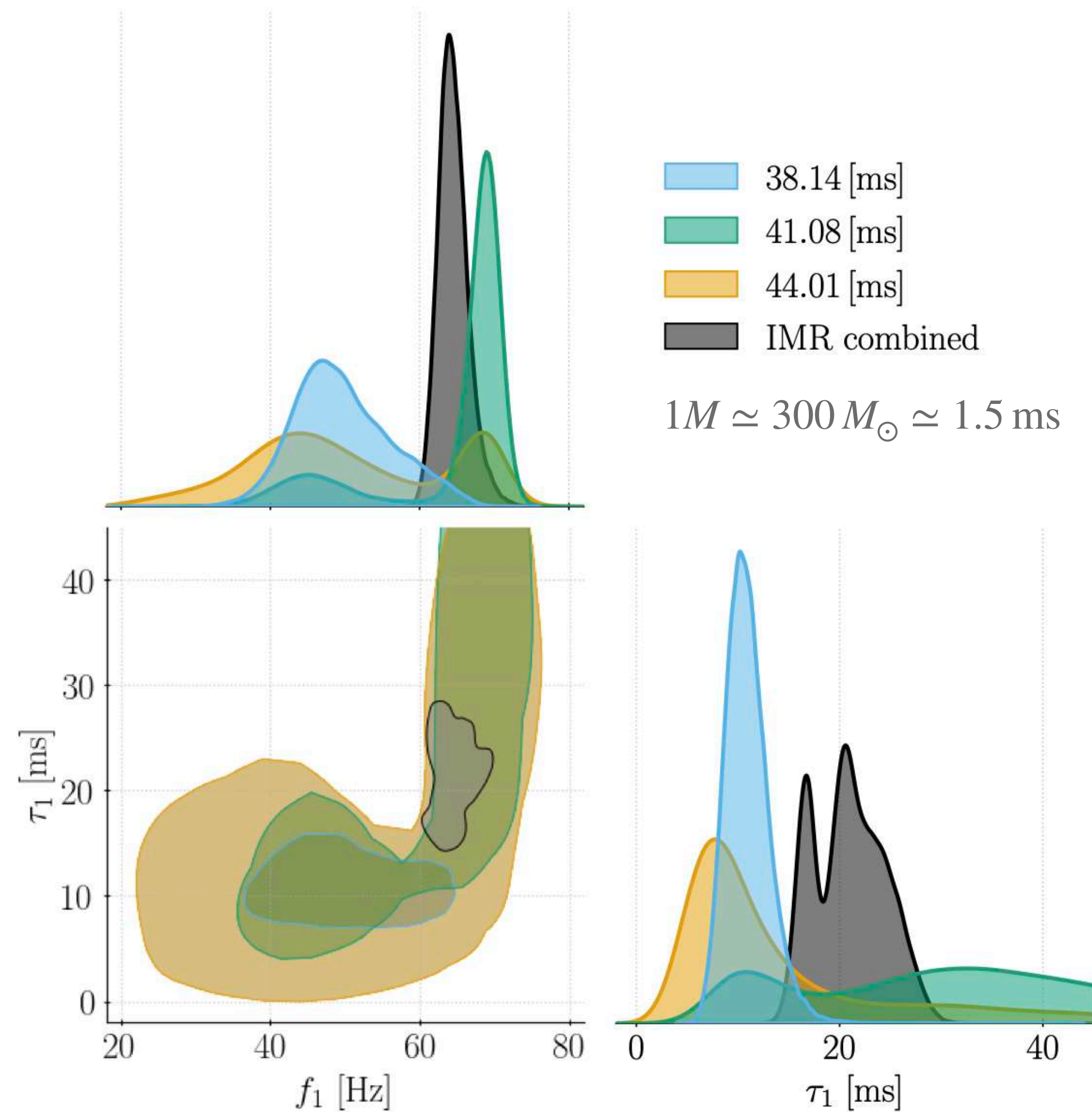
an opportunity and a challenge



$h(t)$

time

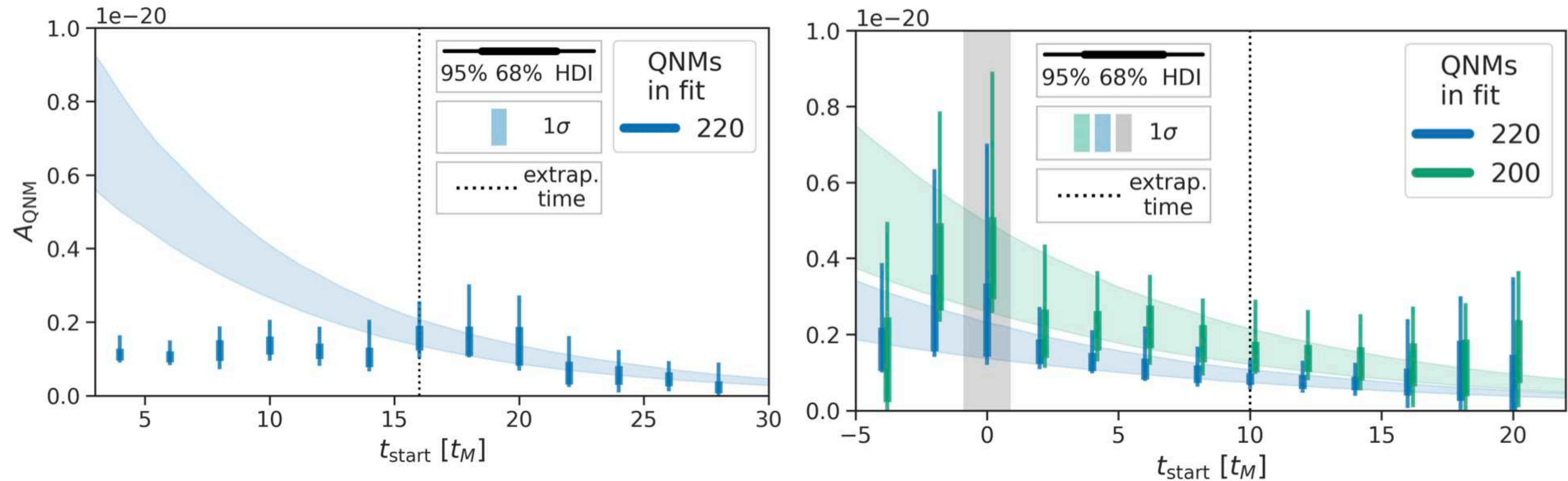
GW231123



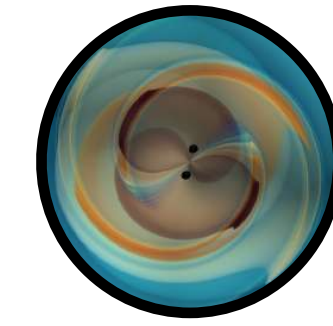
Abac+inc. **M. Isi** [arXiv:2507.08219]

Siegel+(inc. M. Isi) 2025 [arXiv:2511.02691]

GW231123



Siegel+(inc. **M. Isi**) 2025 [arXiv:2511.02691]

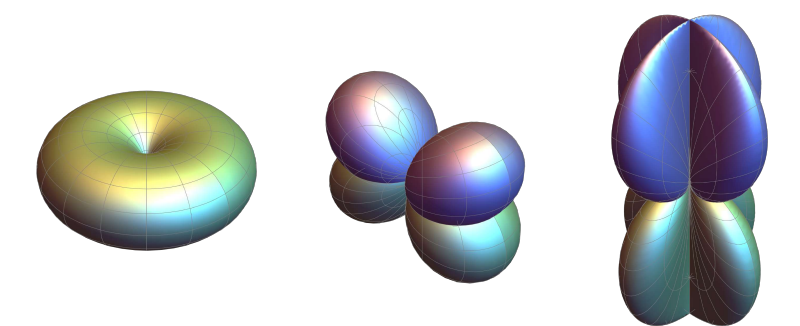


GW190521

the GW190521 signal is merger-ringdown dominated

but the ringdown doesn't look like we naively expect!

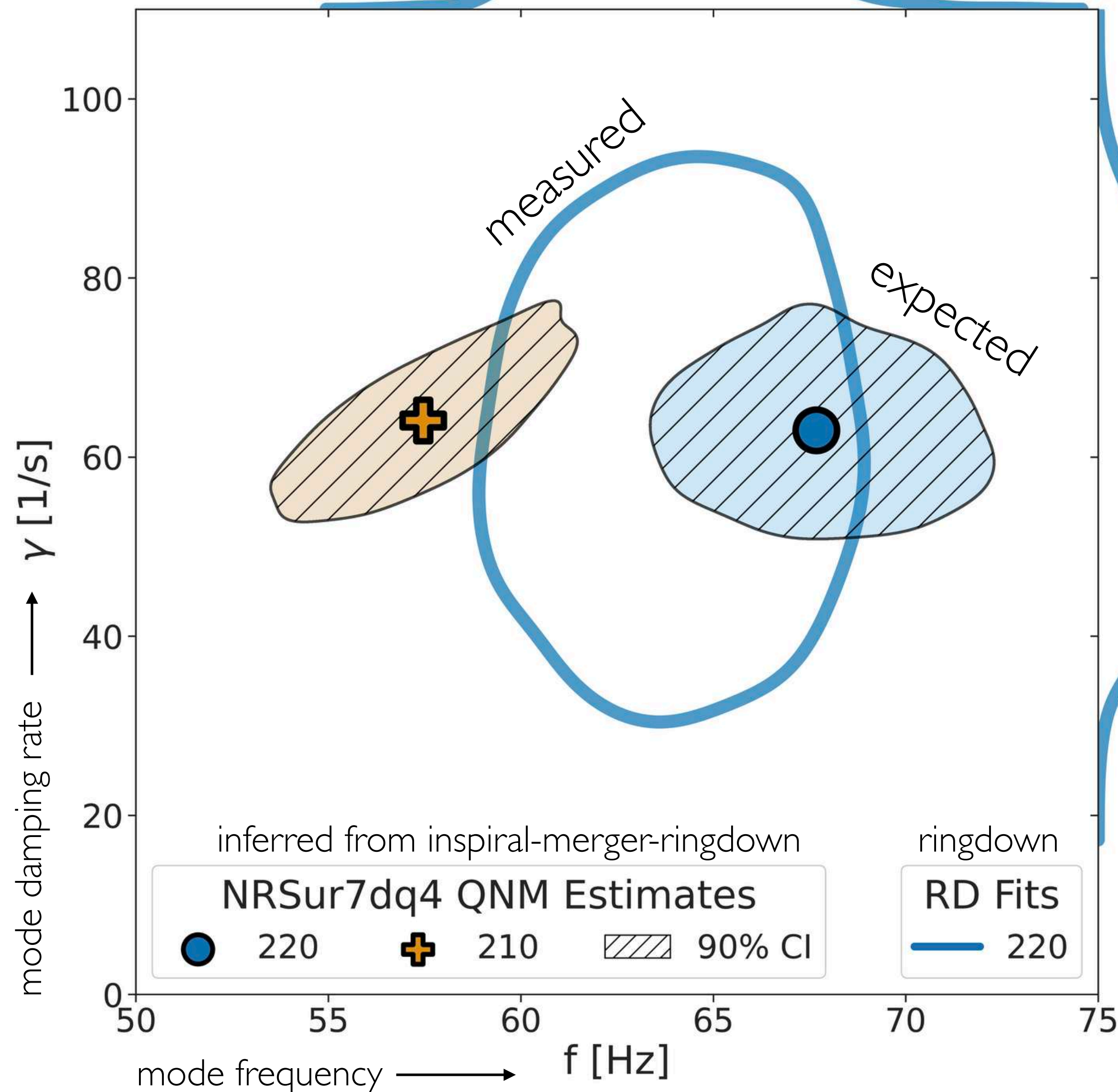
may be evidence of *multiple* angular modes



H. Siegel*, **M. Isi**, W. Farr (PRD 2023)

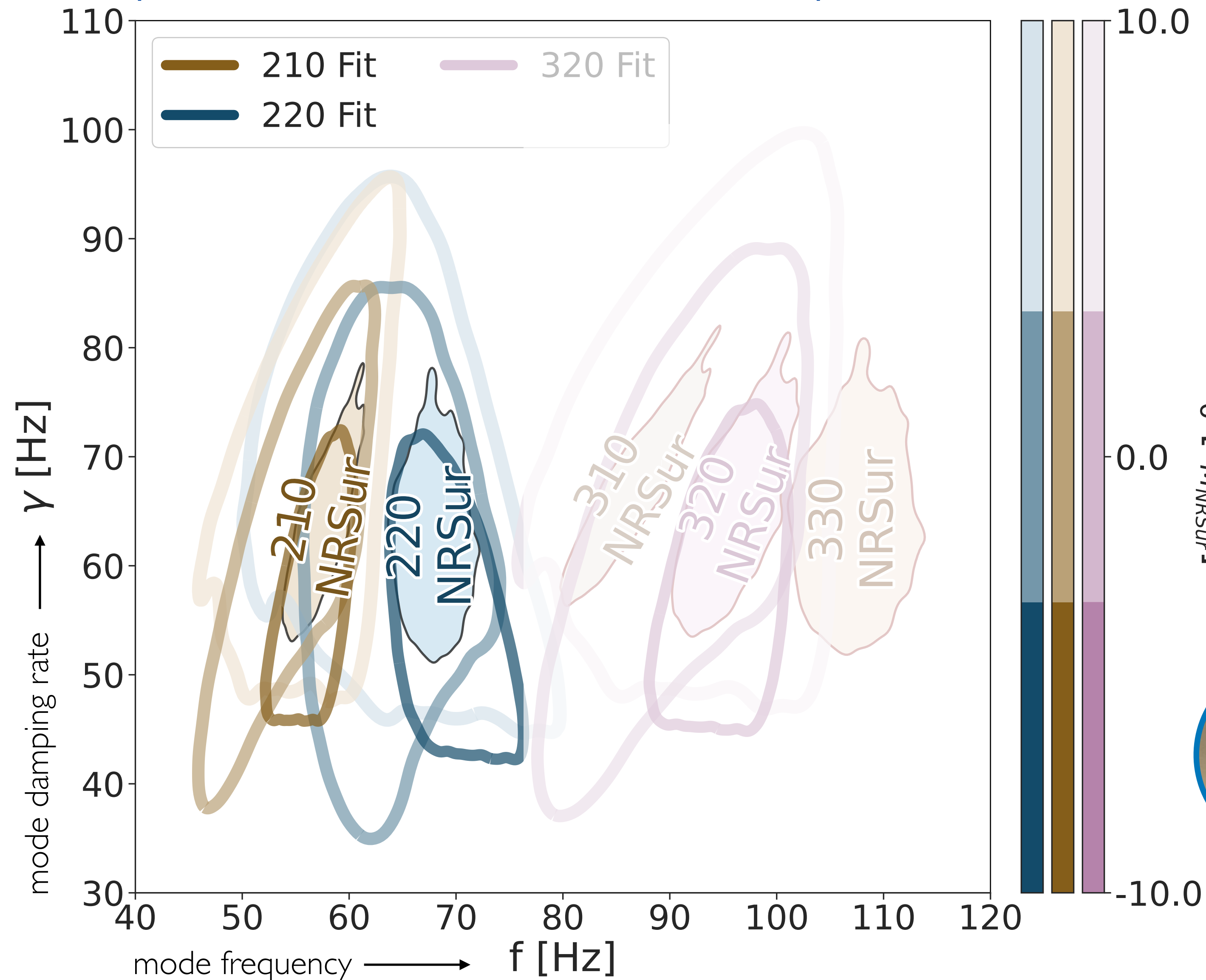
[arXiv:2307.11975]

* student @ Columbia



210 stabilizes fit

preferred w/ nonzero amplitude

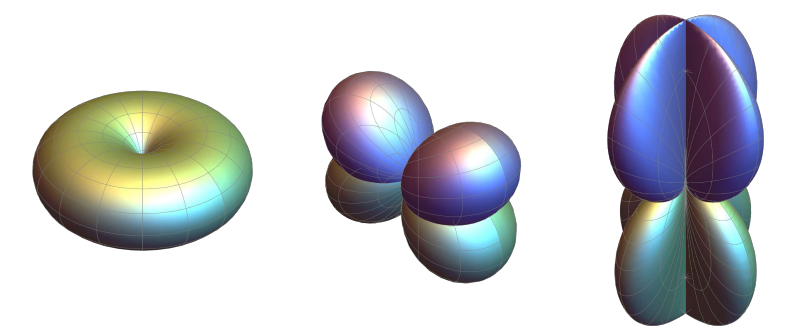


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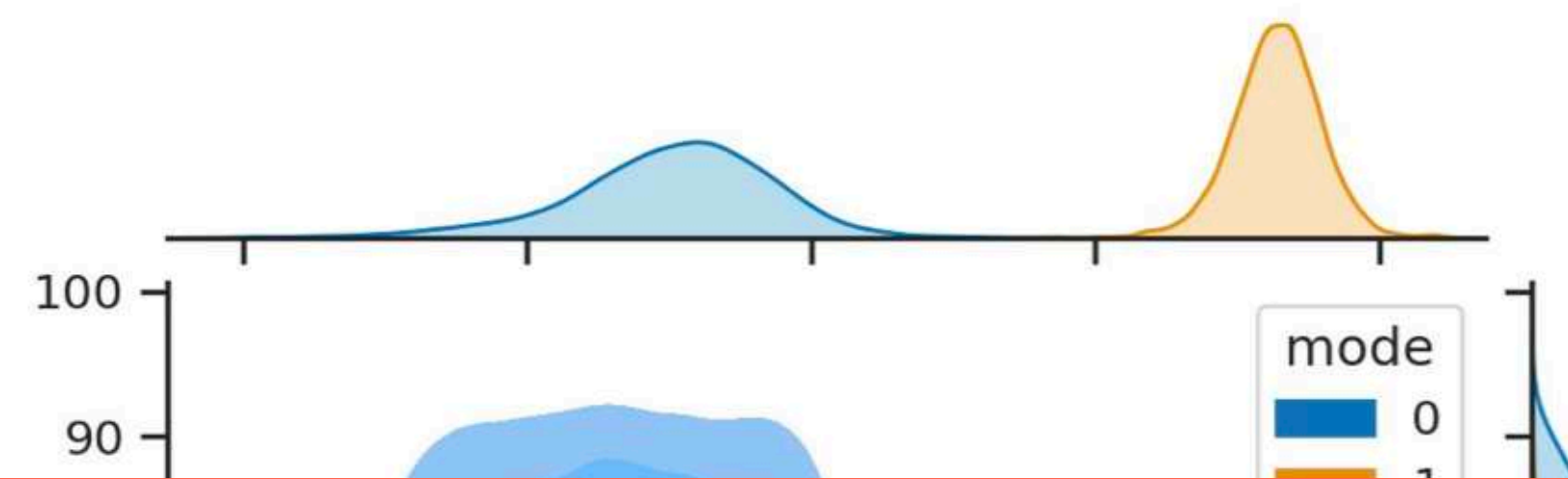


H. Siegel*, **M. Isi**, W. Farr (PRD 2023)

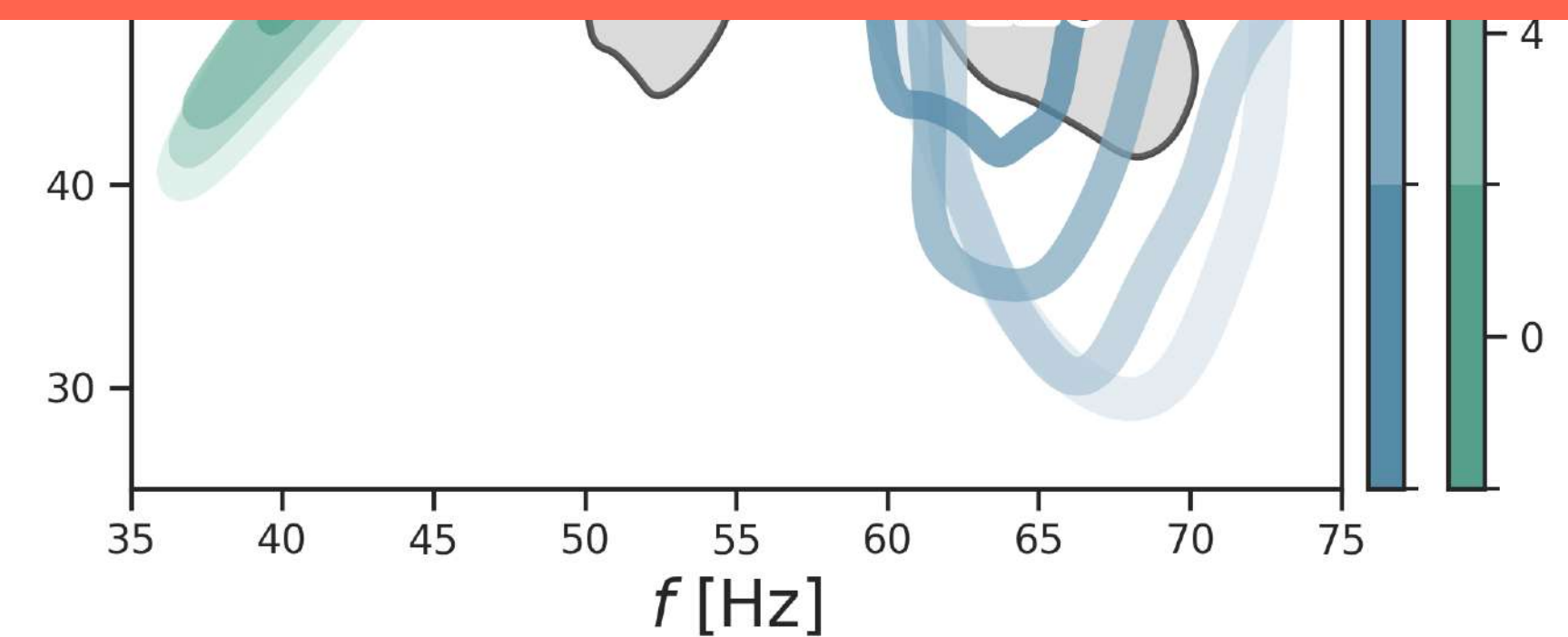
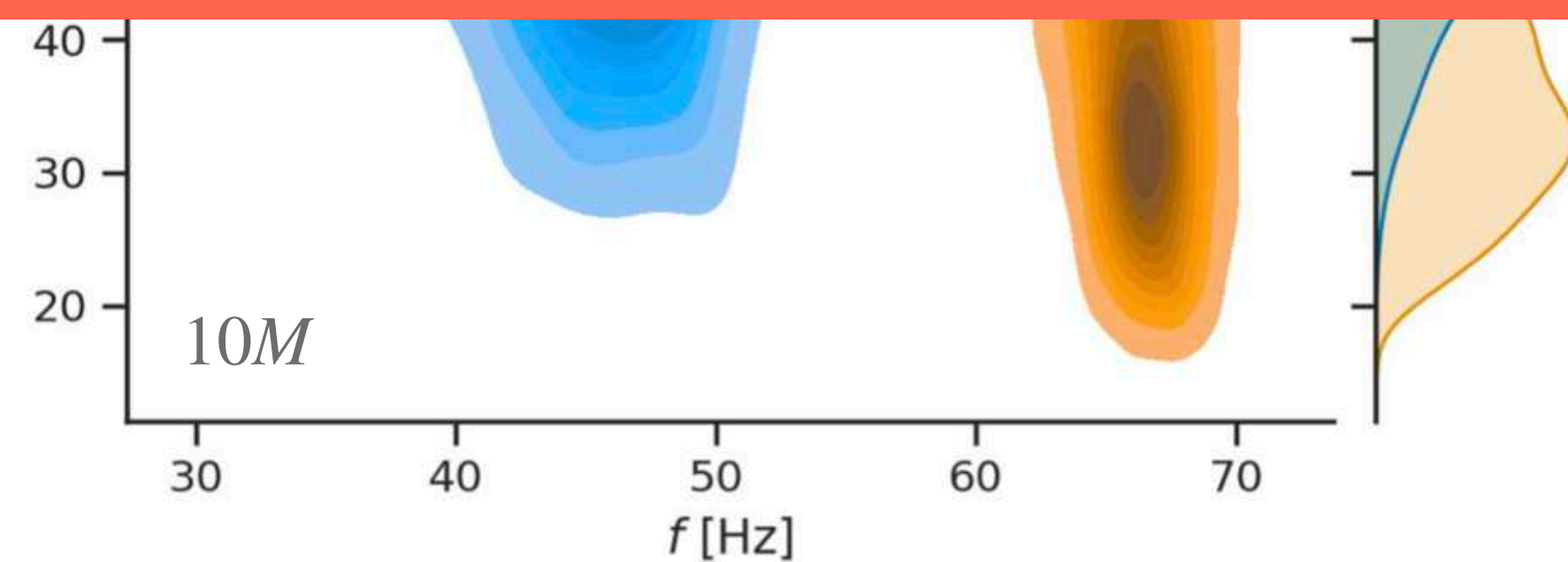
[arXiv:2307.11975]

* student @ Columbia

GW231123



misaligned spins?

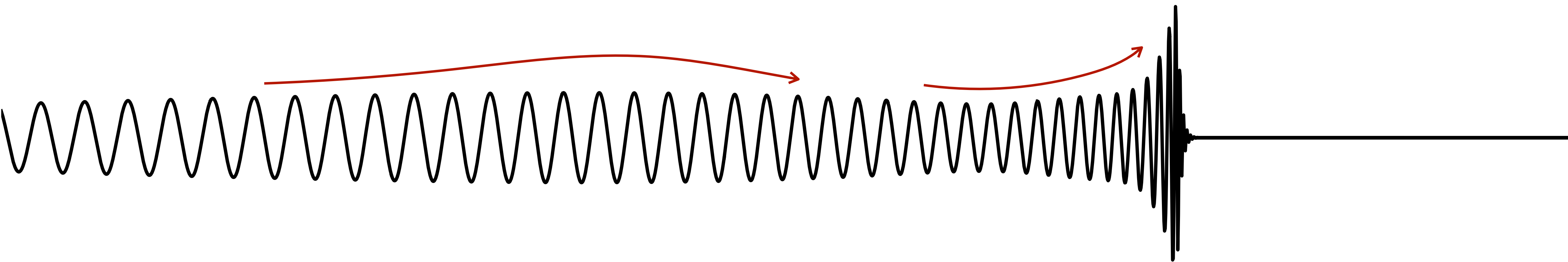


H. Siegel, M. Isi, + [in prep]

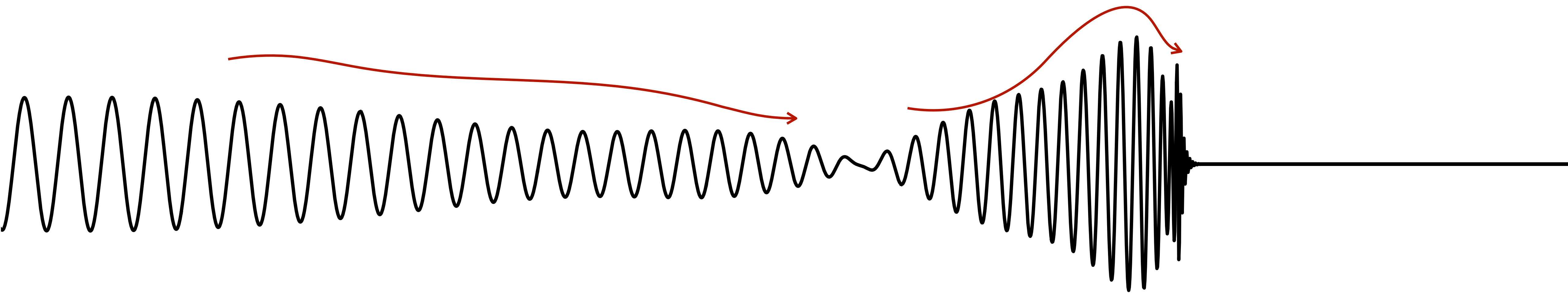
$1M \simeq 300 M_{\odot} \simeq 1.5 \text{ ms}$

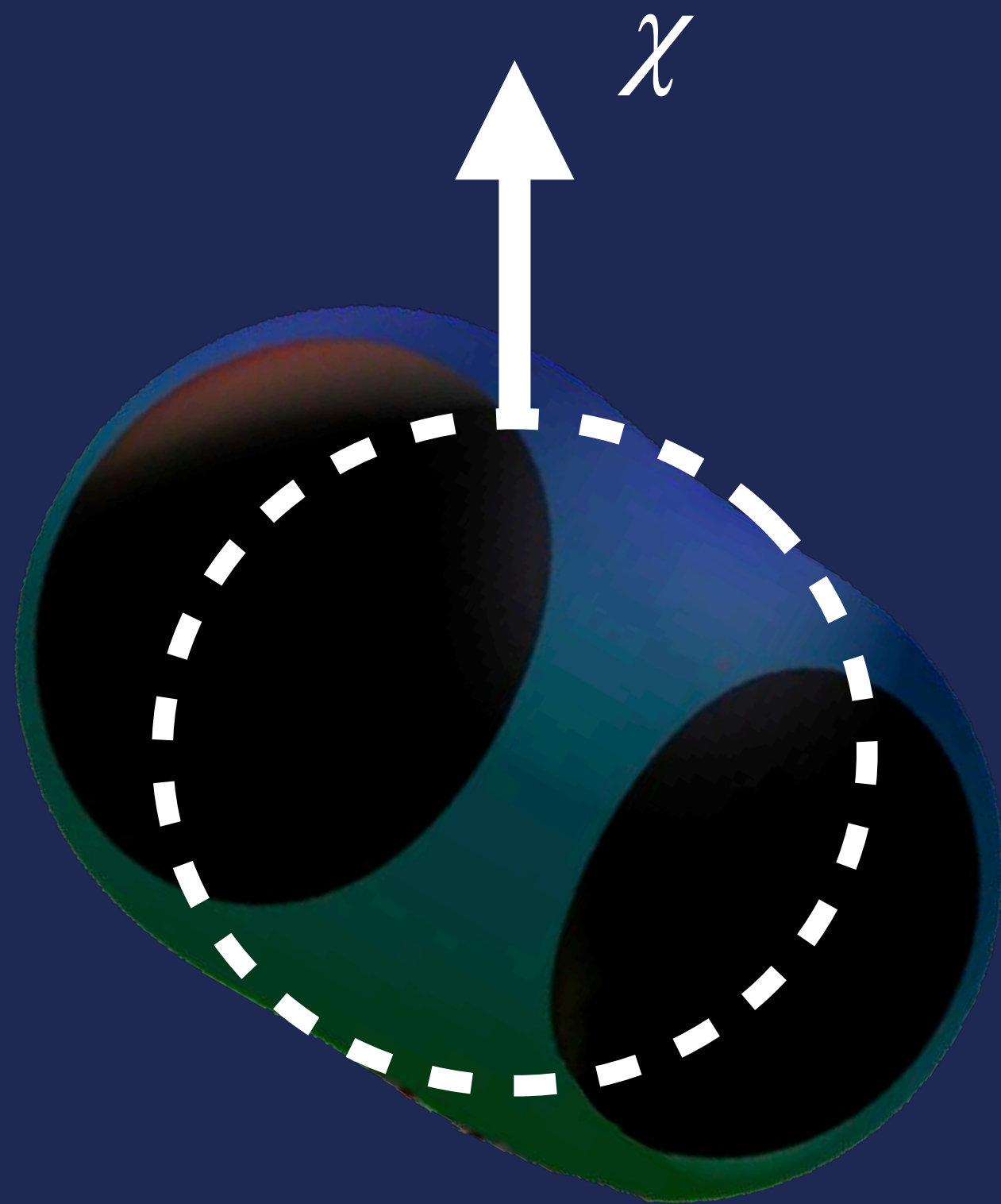
H. Siegel, M. Isi, + [in prep]

precession



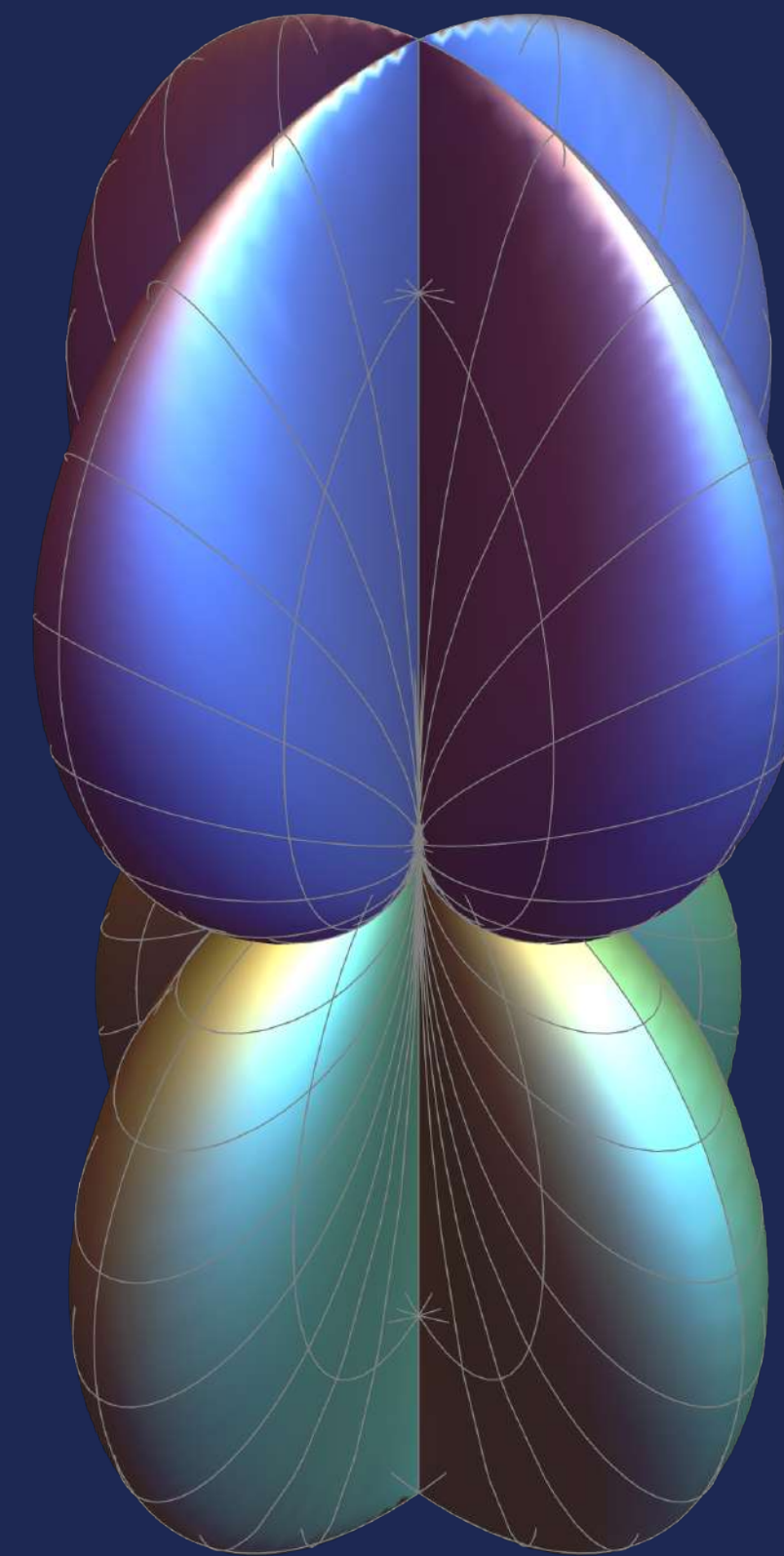
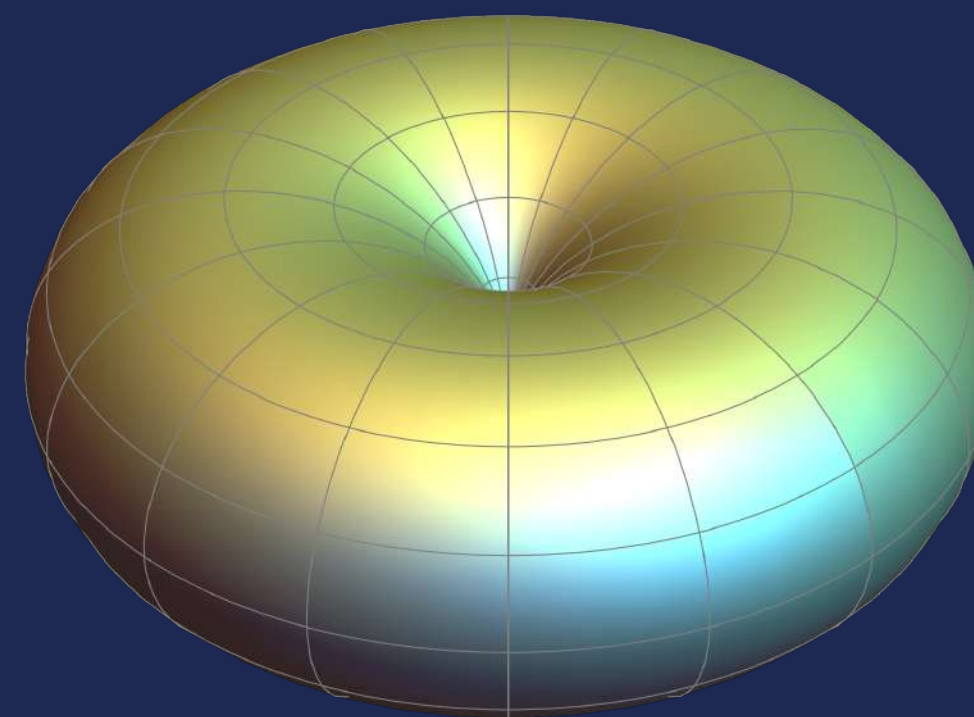
precession



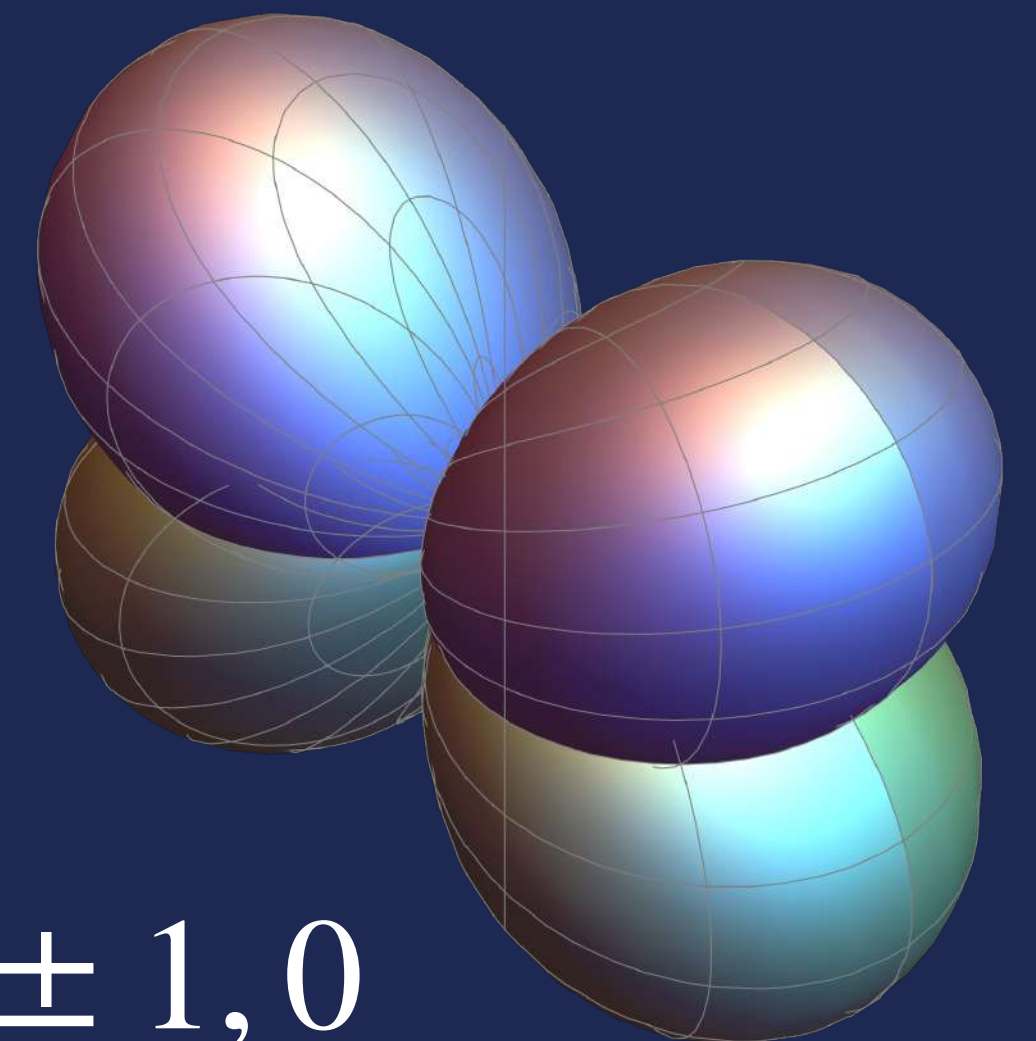


remnant-to-be

quadrupolar
perturbation
misaligned



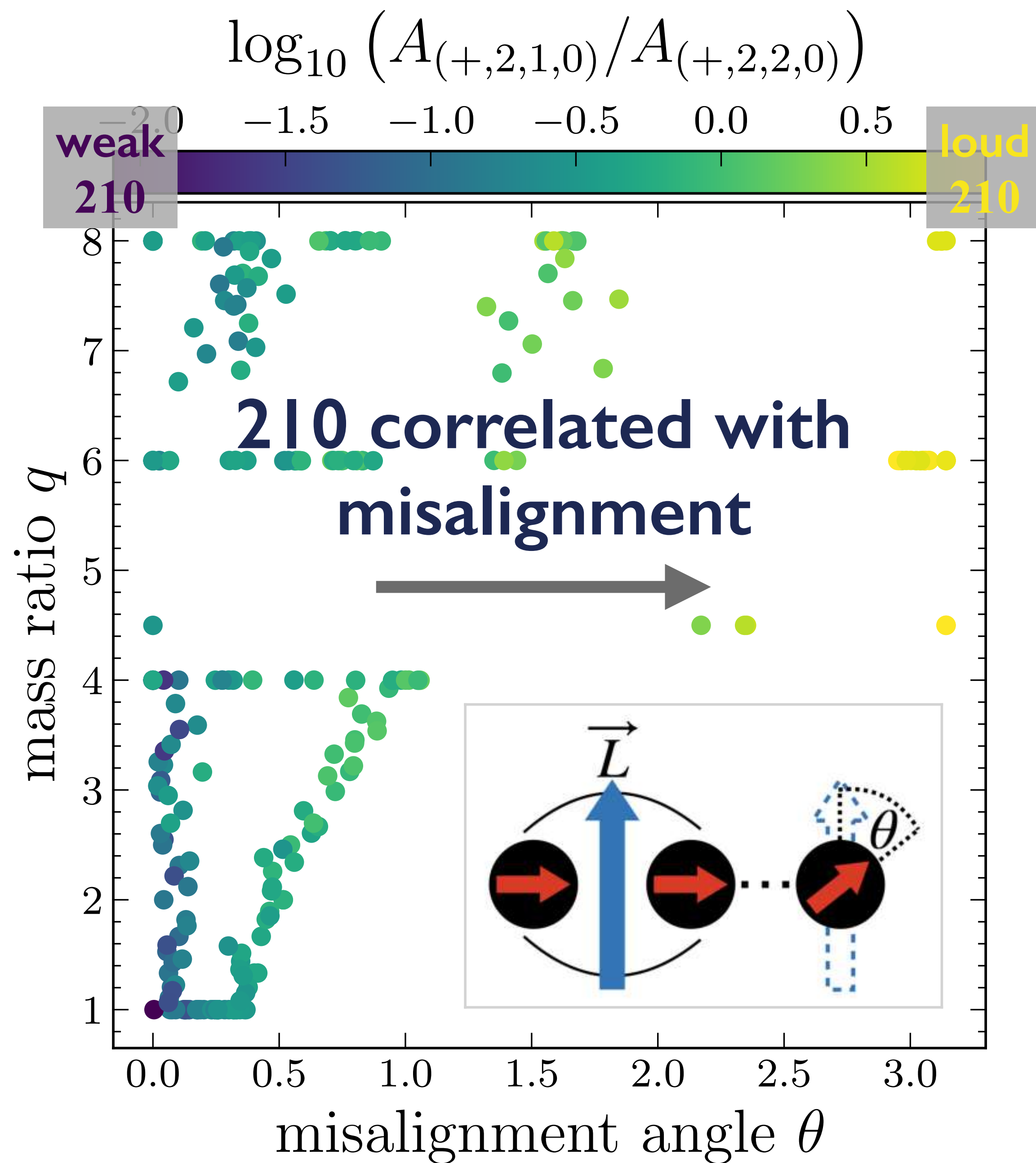
OBSERVABLE!
*different
frequencies*



H. Zhu*, H. Siegel*, K. Mitman*, **M. Isi**, et al
(2023 [arXiv:2312.08588])

*students @ Princeton / Columbia / Caltech

$$\ell = 2, m = \pm 2, \pm 1, 0$$



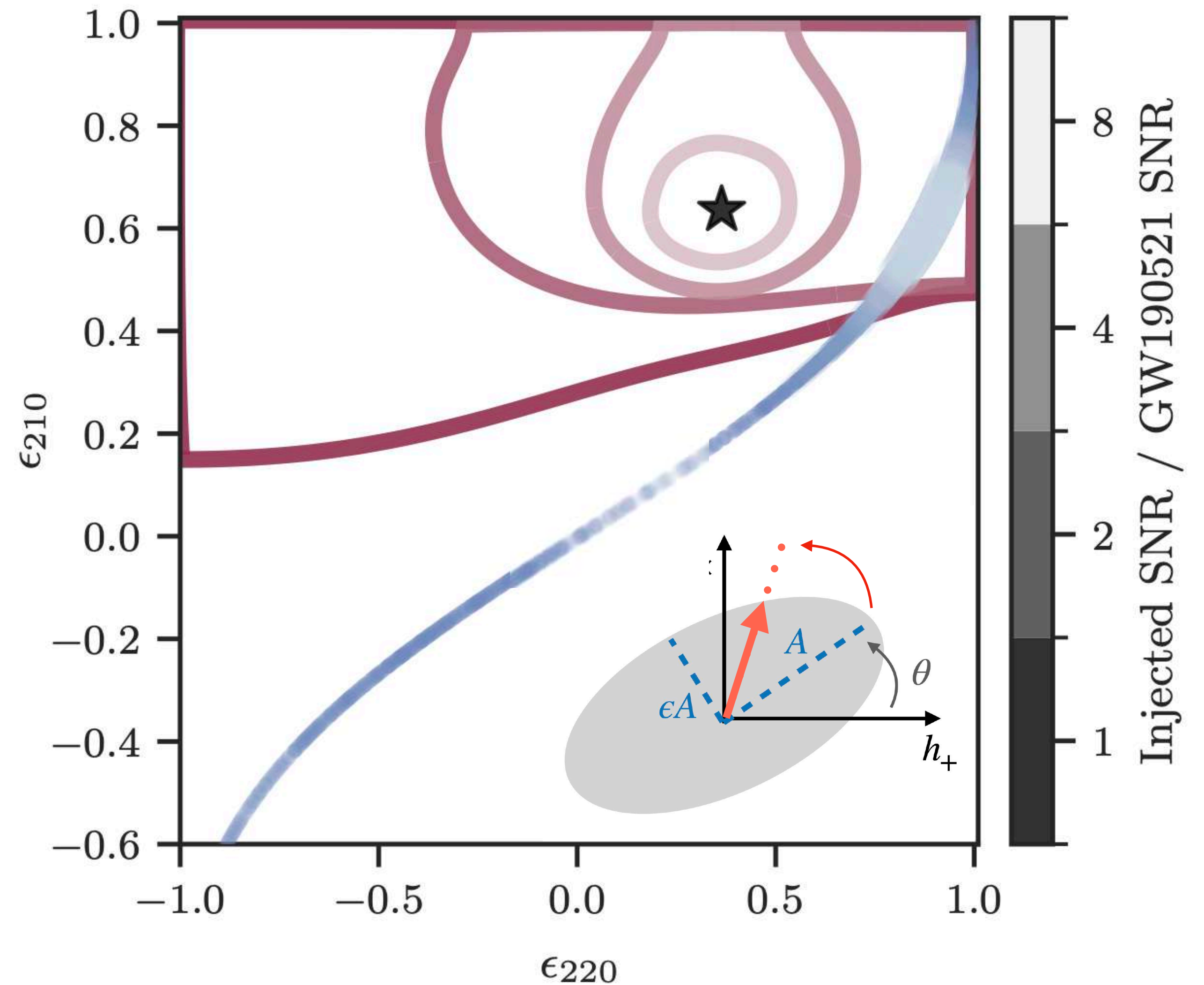
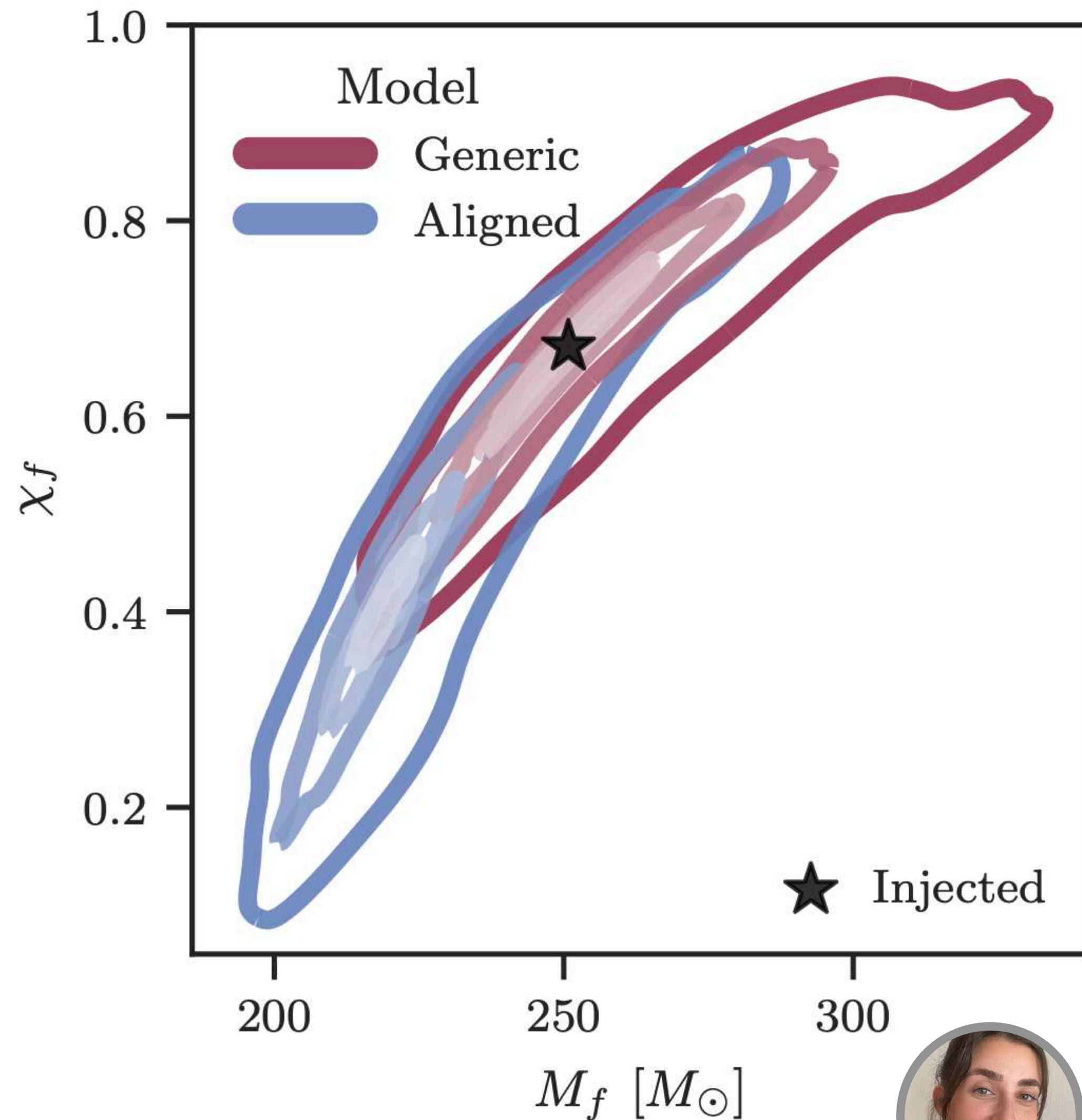
misalignment between
remnant spin and $\vec{L}$ can
excite $l \neq m$ QNMs

study of NR ringdowns with
remnant spin misalignment

H. Zhu*, H. Siegel*,
K. Mitman*, **M. Isi**, et al (2023)
[arXiv:2312.08588]



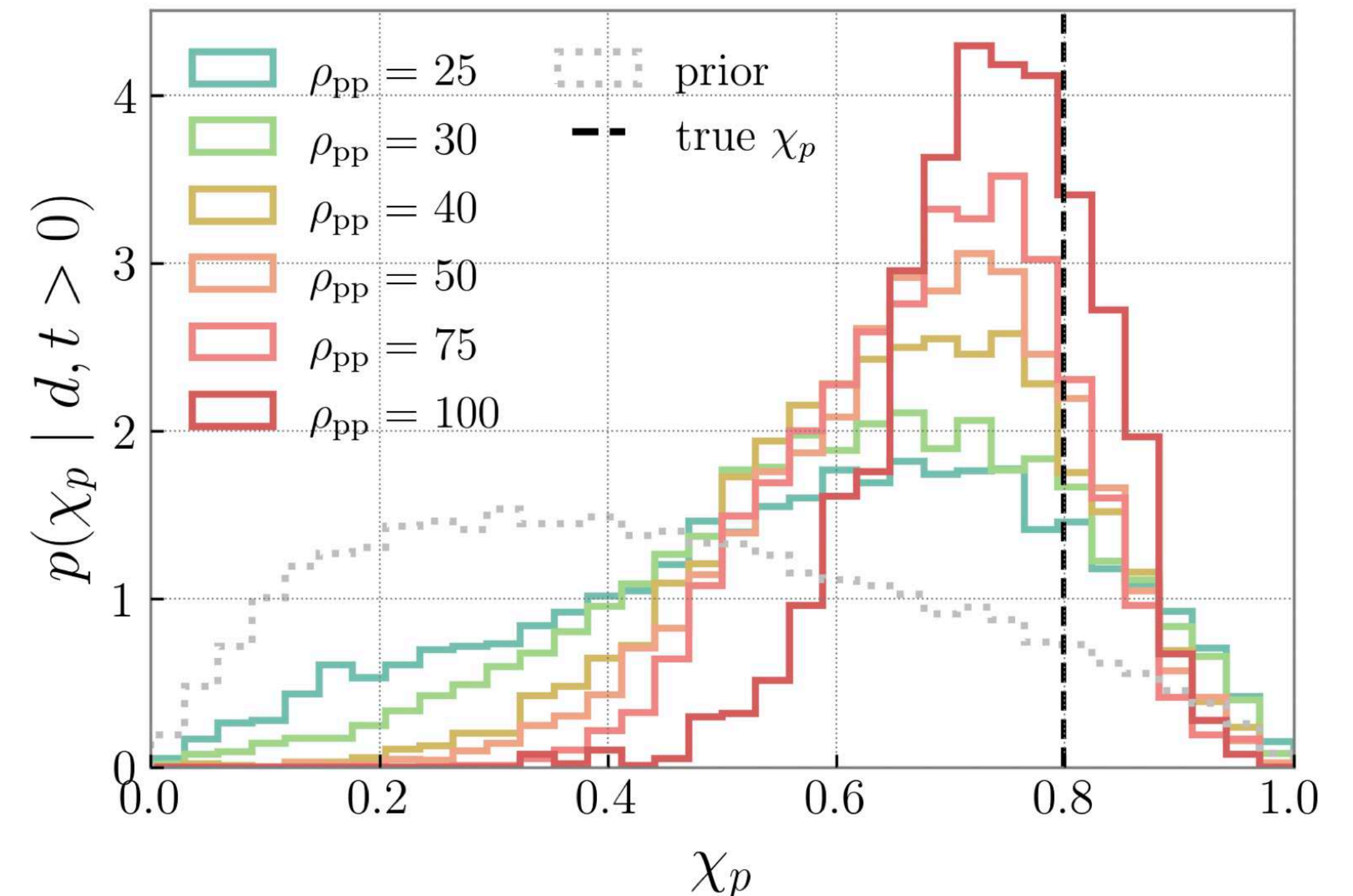
precession imprinted in *polarizations*



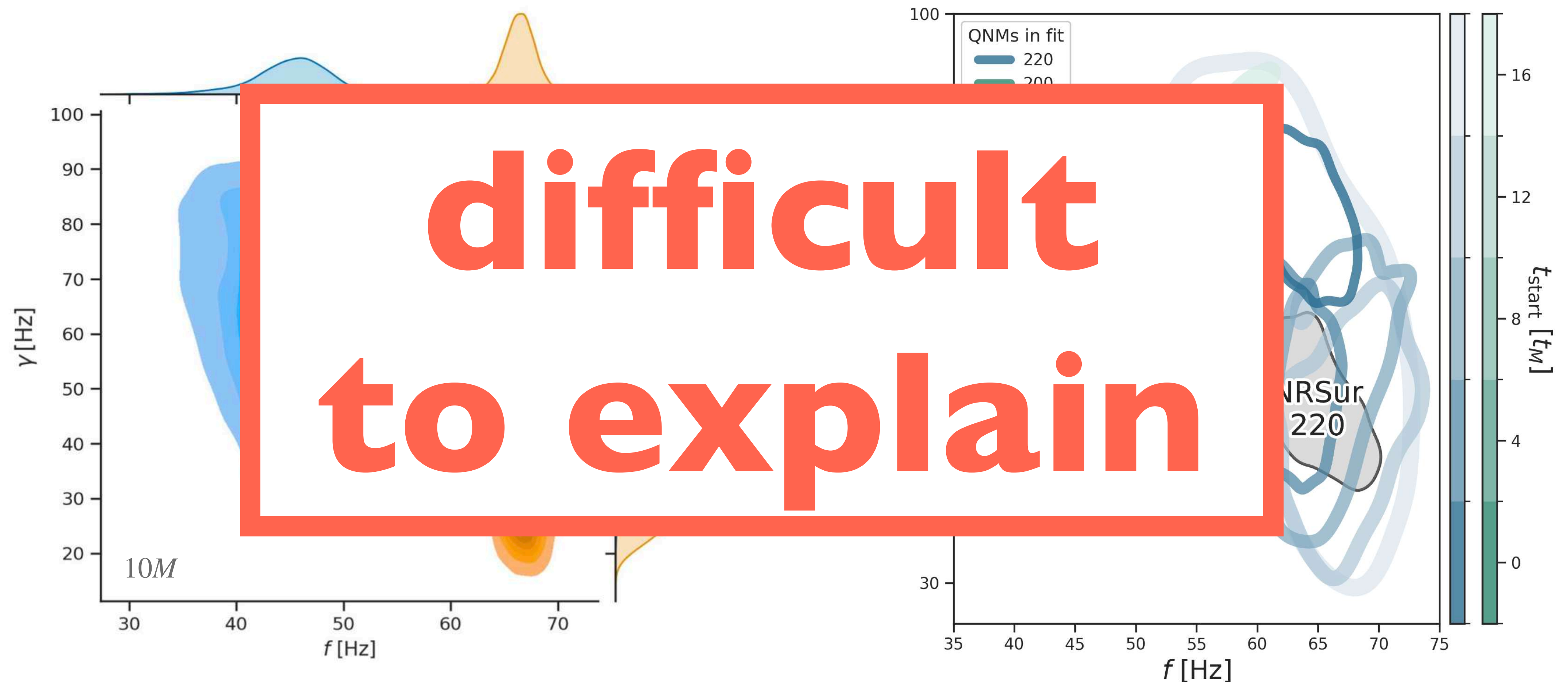
measure precession from ringdown alone

S. Miller, **M. Isi**, + [arXiv:2505.14573]

simulated GW190521-like system with NRSur



GW231123



H. Siegel, M. Isi, + [in prep]

$1M \simeq 300 M_{\odot} \simeq 1.5 \text{ ms}$

H. Siegel, M. Isi, + [in prep]

GW231123 possibilities

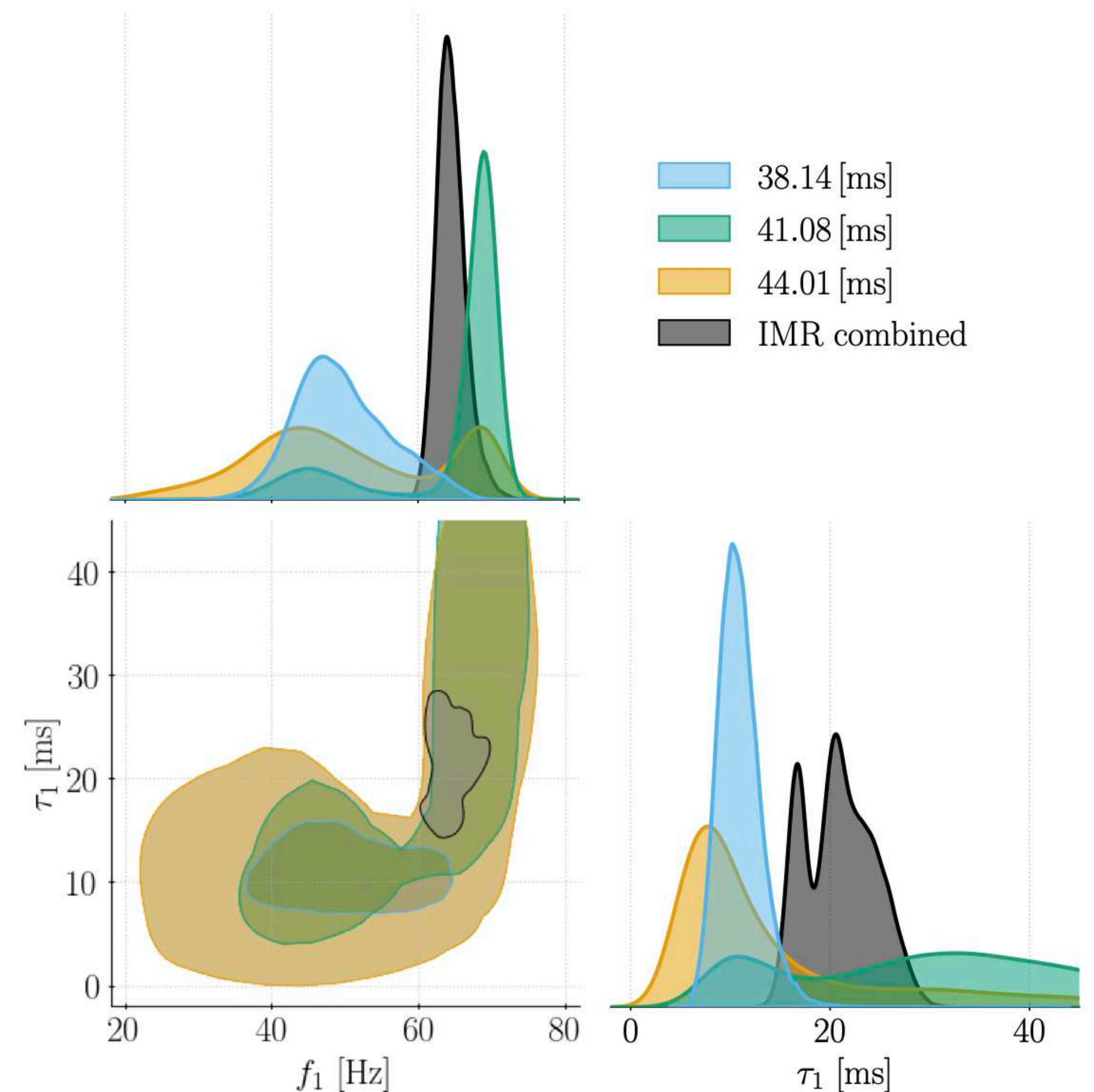


The GW231123 signal remains difficult to explain morphologically.

Options:

1. GW231123 is very special BBH
2. GW231123 is not a BBH
3. All our waveforms are quite wrong

Either way, need better *understanding* of the merger process



Abac+inc. M. Isi [arXiv:2507.08219]

OUT THERE

These Black Holes Shouldn't Exist, but There They Are

On the far side of the universe, a collision of dark giants sheds light on an invisible process of cosmic growth.

Science / Space

Astronomers detect most massive black hole collision to date

By [Jacopo Prisco](#), CNN

🕒 7 min read · Updated 11:00 AM EDT, Thu July 17, 2025



This illustration depicts two black holes colliding in space. The collision sends gravitational waves, ripples in the curvature of spacetime, radiating through space. These waves can be detected by observatories on Earth.

ILLUSTRATION BY MARK GARLICK, SCIENCE PHOTO LIBRARY

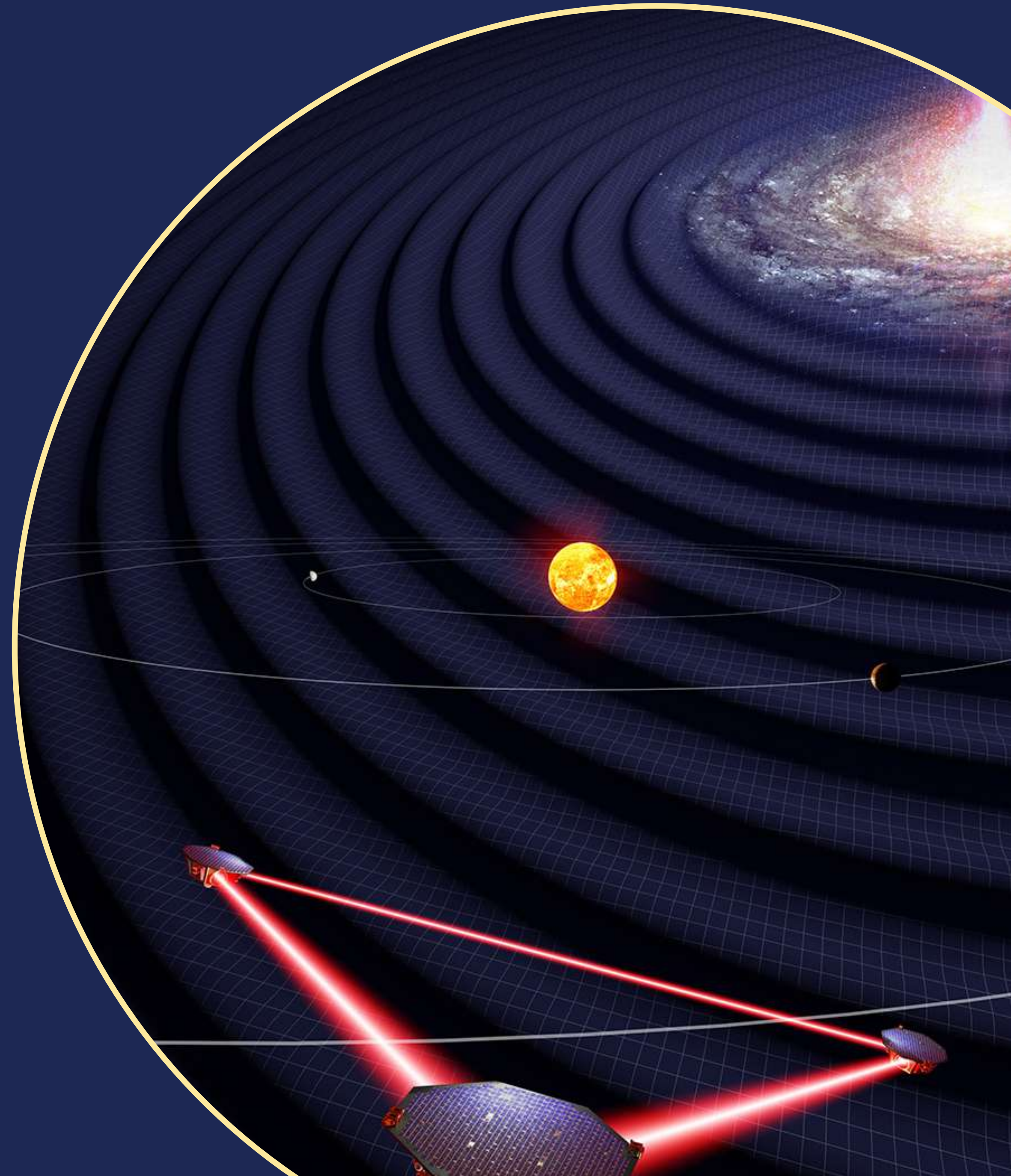
SCIENCE

Scientists record a black hole collision they weren't sure was possible

The largest black hole collision ever recorded has scientists' jaws on the floor — and scratching their heads.

conclusion.

looking to
the future



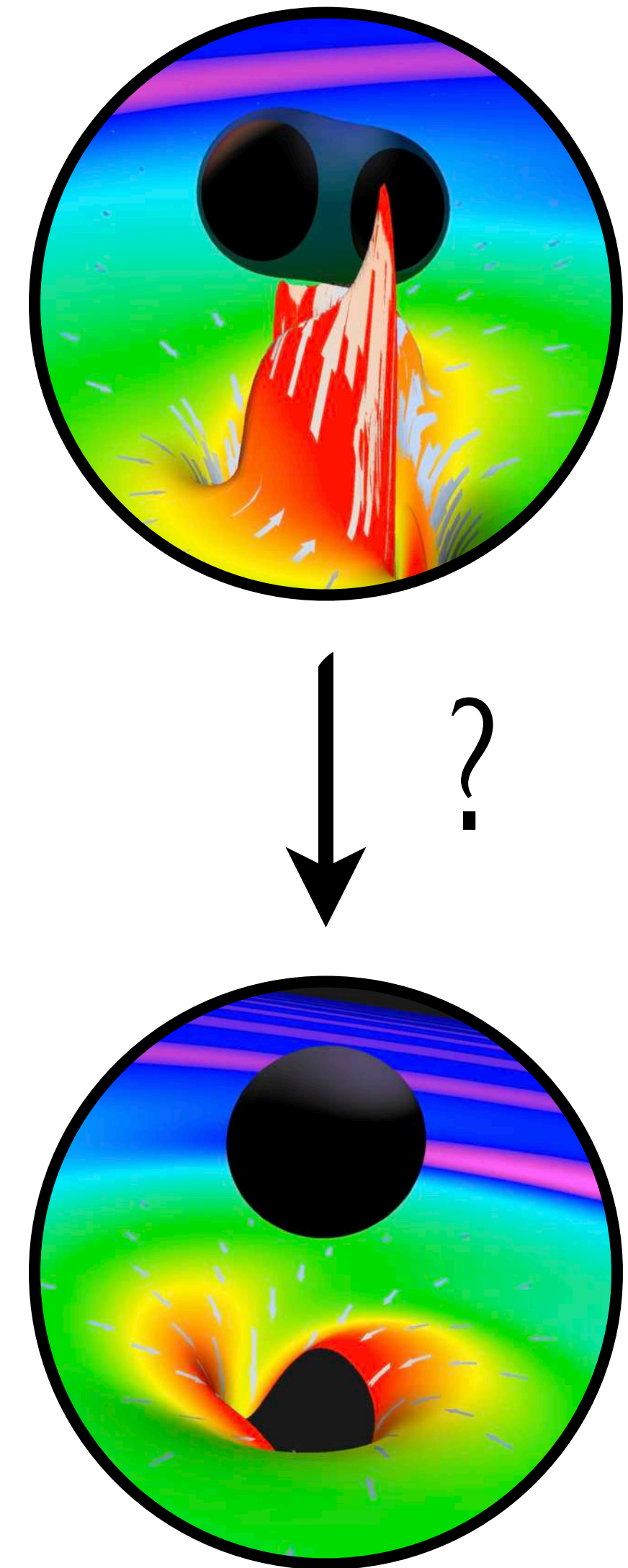
parting thoughts

GW observations are **pushing our understanding** of merger physics

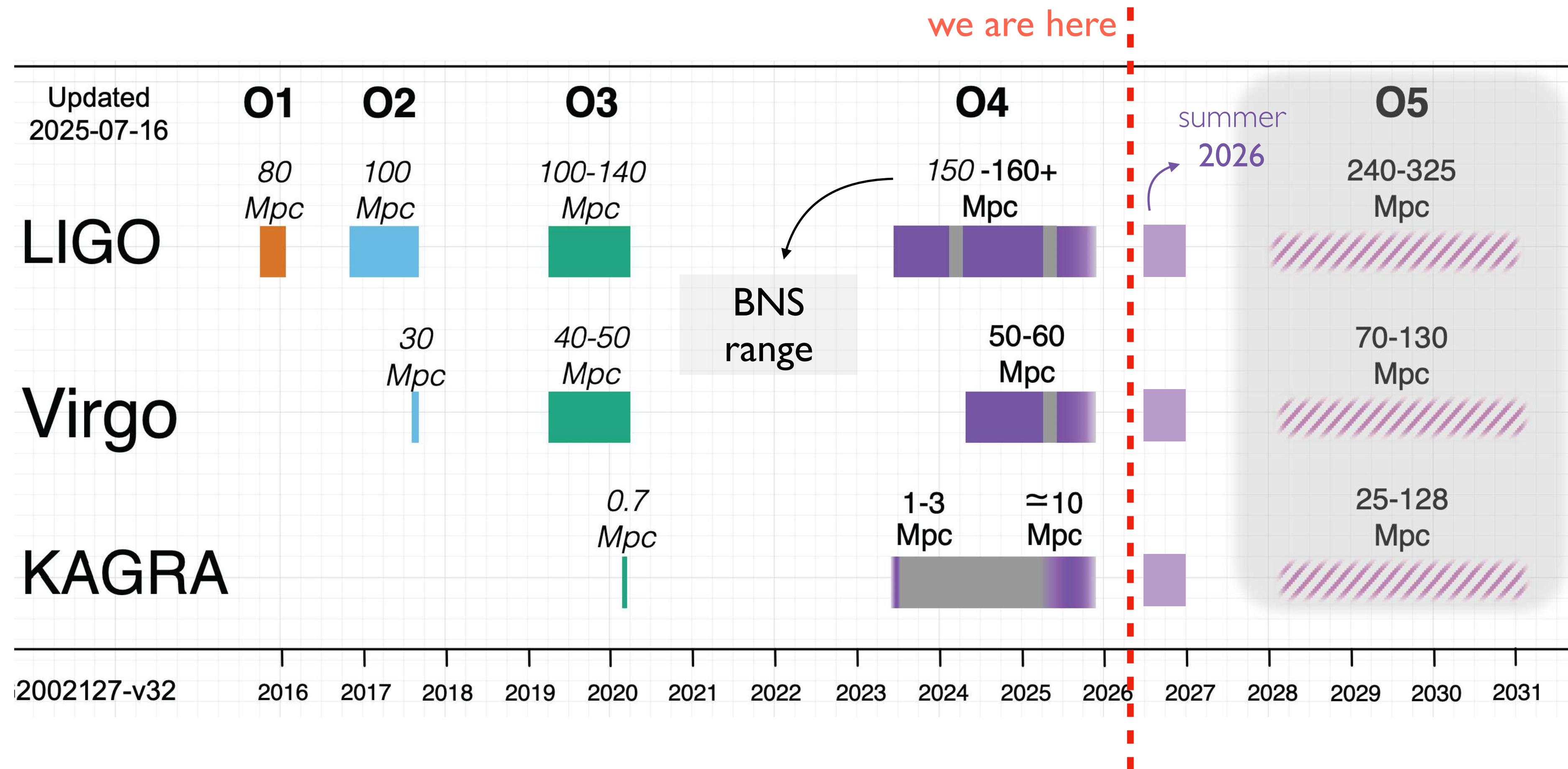
very little has worked out as had traditionally been expected

analytic intuitions still have a big role to play!

important for **astrophysics** too



observation timeline



<https://observing.docs.ligo.org/plan/#timeline>
https://gwosc.org/detector_status/

